The Daily Star

Presents

# AI For All

In Association With

AI FORUM BANGLADESH

UNITED INTERNATIONAL UNIVERSITY

IRIIC

SPEAKER

## PROF. KHONDAKER A. MAMUN, PHD.

Professor of AI and Digital Transformation,
Department of CSE,
United International University (UIU)

26 April 2026

The Daily Star Center

# AI FOR DEVELOPMENT

## EMPOWERING YOUTH AND TRANSFORMING EQUITY


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Founder Director, Advanced Intelligent Multidisciplinary Systems (AIMS) Lab, UIU

Founder Director, Brain Computer Interface Research Lab (BCI Research Lab), UIU

Founder Director, Institute of Research, Innovation, Incubation & Commercialization (IRIIC), UIU

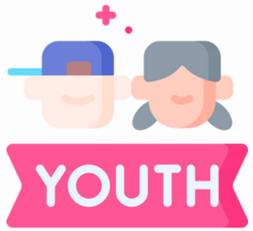
Founder, CMED Health (An AI for Health initiatives towards UHC of SDG)

Founder, AI Forum Bangladesh (Responsible AI for All)

Vice President - AI, Innovation and Entrepreneurship at Washington University of Science and Technology, USA

UNITED INTERNATIONAL UNIVERSITY · IRIIC · AIMS Lab · AI FORUM BANGLADESH · BUILDING INNOVATION ECOSYSTEM · Career Care · AI ACADEMY · SUSASTHO · SUSASTHO.AI · CMED

# Bangladesh at a Turning Point




**Youth-majority nation at a historic crossroads**

**Economic pressure, climate stress, global uncertainty**

**Digital penetration rising, trust in systems declining**

**Youth no longer passive recipients of policy**

# The Youth Reality

**Over 50% of the population is under 30**

**Aspirational, digitally connected, politically aware**

**High energy, low institutional confidence**

**Talent abundant, pathways constrained**

# Youth and Inequality

Youth wants opportunities not inequalities

# Inequality in Education

Education quality is determined more by postcode than potential

## Urban

**Access to quality schools & universities**

**Access to labs, libraries, internet**

**Clear pathways to higher education & jobs**

## Rural

**Quality Teacher shortages & overcrowded classrooms**

**Irregular classes, limited subject teachers**

**Bright students drop out after SSC/HSC**

# Inequality in Health

Survival chances depend on location, not medical need

## Urban

Hospitals and clinics equipped with specialists and diagnostics nearby

Emergency services readily available

Regular checkups and  care accessible but Costly

## Rural

Limited doctors, clinics often understaffed and under-resourced

Long travel required for basic treatment; higher costs

Care delayed, survival depends on location rather than medical need

# Inequality in Agriculture

Bangladesh's agriculture is creating a new form of inequality

## Urban

**Access to modern machinery, high-yield seeds, and technology**

**Easy access to finance, markets, and supply chains**

**Agri-business opportunities and training programs available**

## Rural

**Small farmers lack machinery, improved seeds, and irrigation**

**Limited access to credit and markets; middlemen dominate**

**Innovation and training rarely reach villages**

# Inequality in Public Service

Rights exist on paper, access exists only in cities

## Urban

**Easy access to the government office, banks**

**Digitized systems & service centers**

**Information is easier to find**

## Rural

**Long travel for birth certificates, NID, and land records**

**Manual processes, unclear procedures**

**Dependence on middlemen**

# Inequality in Environment

Rural people face climate disasters with no safety net

## Urban

**Better drainage & waste collection**

**Water supply systems available**

**Environmental risks managed by infrastructure**

## Rural

**Poor sanitation increases disease**

**Floods, river erosion, and salinity destroy homes & crops**

**Climate shocks directly affect income**

# 2024 Youth-Led Uprising

- Quota reform movement led by students
- Triggered by perceived injustice and exclusion
- Rapid digital mobilization across campuses and cities
- Youth demanding fairness, dignity, and voice

## What 2024 Revealed?

- Youth understand systems and power
- Frustration is structural, not emotional
- Merit, fairness, and transparency matter deeply
- Silence has been replaced by collective action

# Living in the AI Era: Embracing the Future

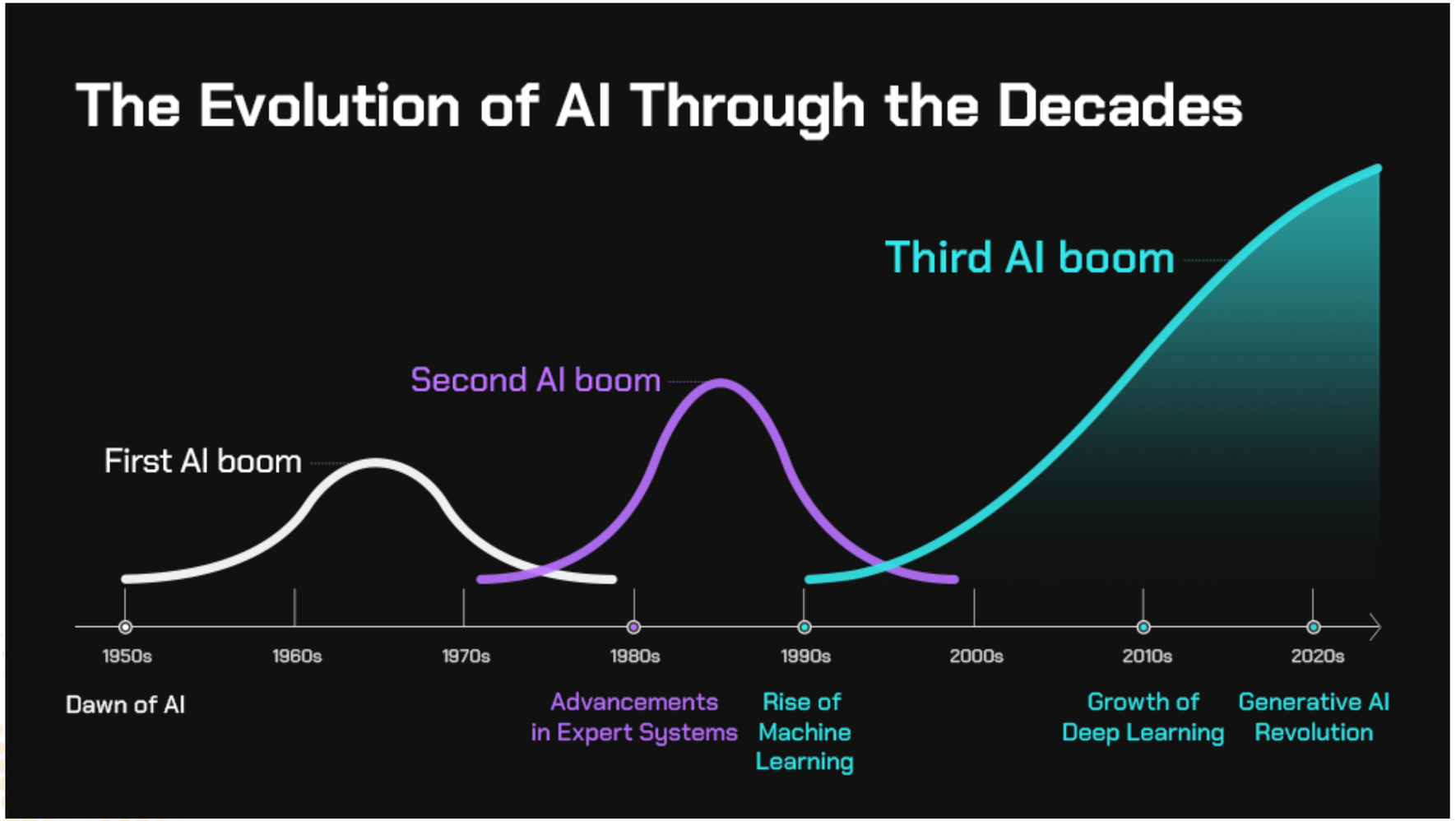
# WHAT IS AI?

## Artificial intelligence (AI)

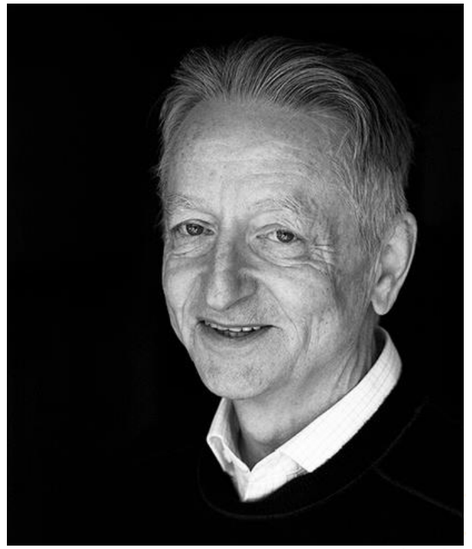
is when we teach computers to think, learn, and make decisions like humans. It's like giving machines a brain.

## From voice assistants to autonomous vehicles AI is transforming everyday life.

# HISTORY & GROWTH OF AI

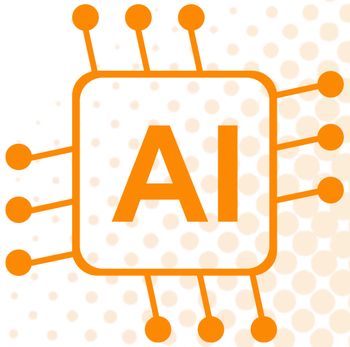



Source: [All About AI] The Origins, Evolution & Future of AI

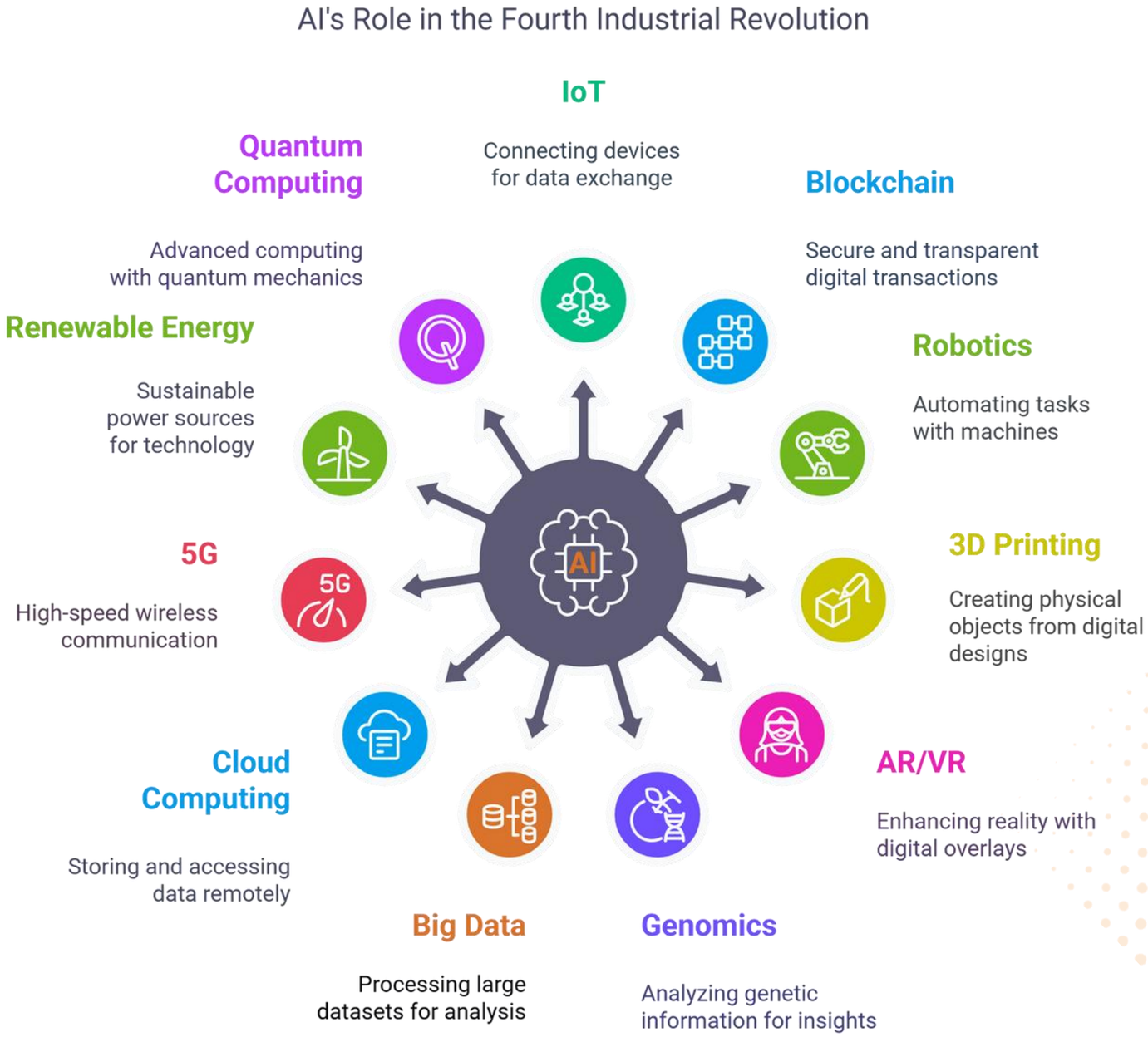


**Geoffrey E. Hinton**

**Nobel Prize in Physics' 2024**

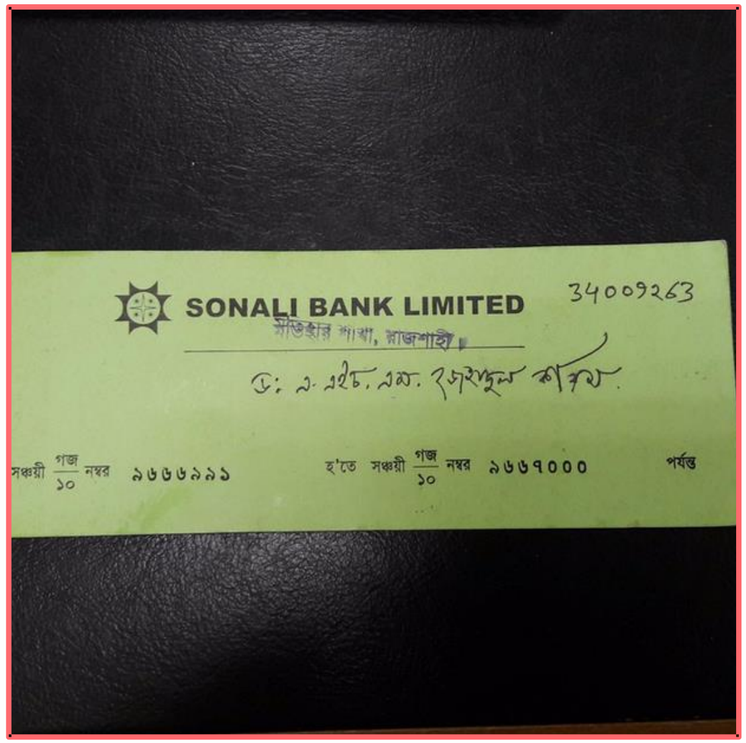


# AI AND THE FOURTH INDUSTRIAL REVOLUTION (4IR)




**AI IS THE CORE TECHNOLOGY DRIVING THE 4TH INDUSTRIAL REVOLUTION**

# DIGITAL TRANSFORMATION




**One Day/Half Day**

**Manual Banking: Long queues, paperwork, and time-consuming transactions.**

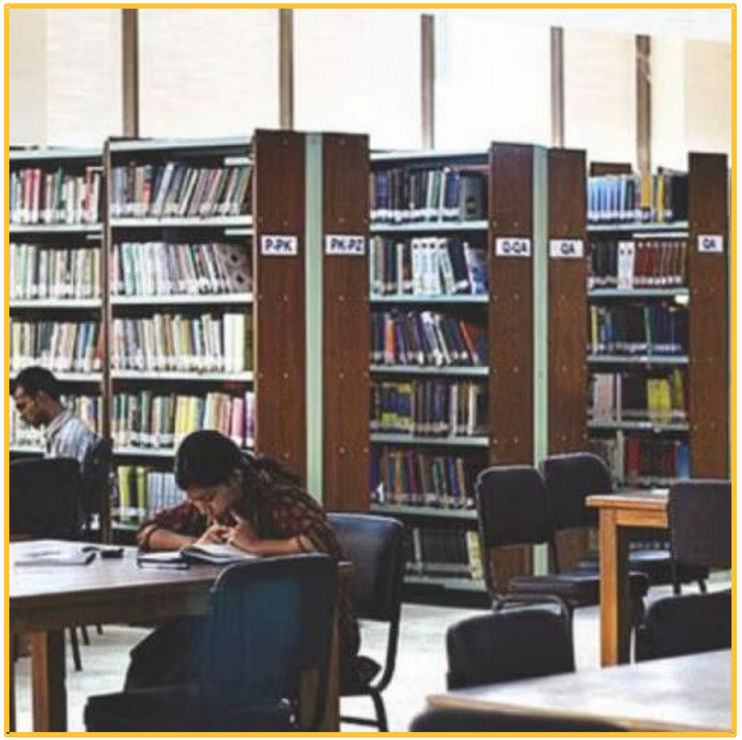
**5 Minutes**

**ATM Era: 24/7 cash withdrawal and balance inquiries through self-service kiosks**

**Less then 1 minute**

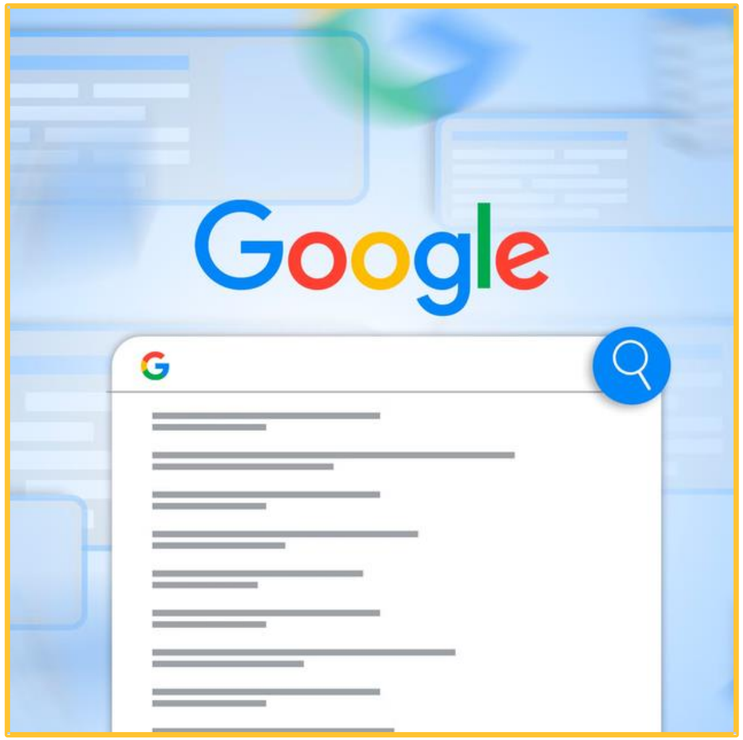
**Mobile Financial Services (MFS): Instant money transfers, utility payments, and digital wallets from your phone.**

# AI TRANSFORMATION



**8 Hours**

**Traditional Library:**
**Hours spent searching through books for information.**




**2 Hours**

**Search Engine Revolution: Google made information accessible with a click**

**Less then 1 minute**

**AI-Powered Assistant: ChatGPT delivers instant, context-aware answers and assignment help.**

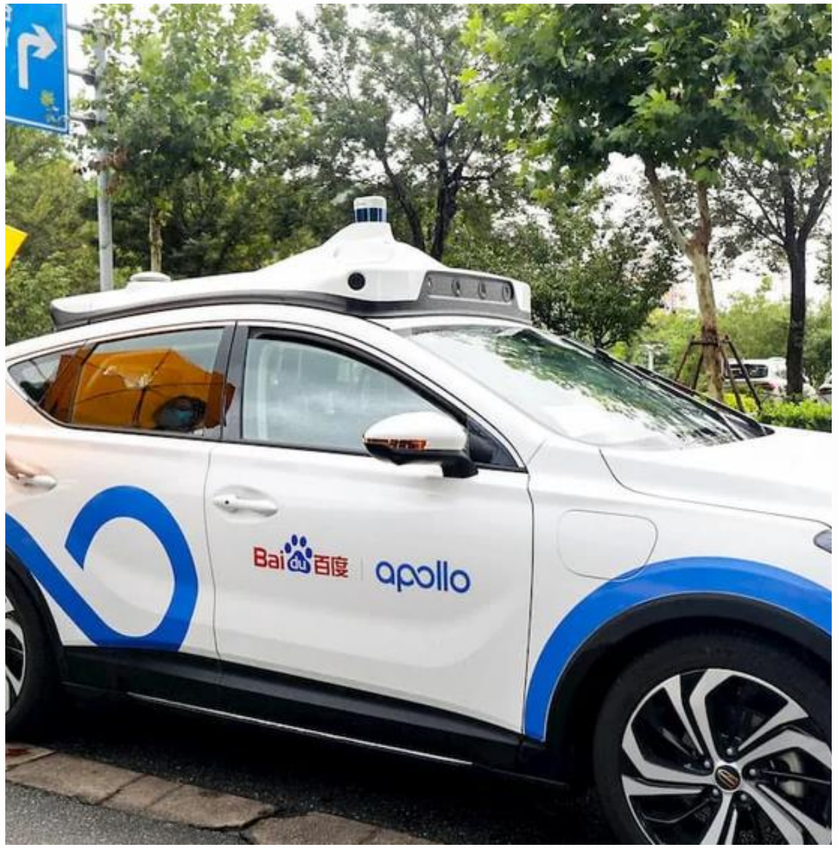
# WHAT IS HAPPENING GLOBALLY?

## Driverless Car

TESLA

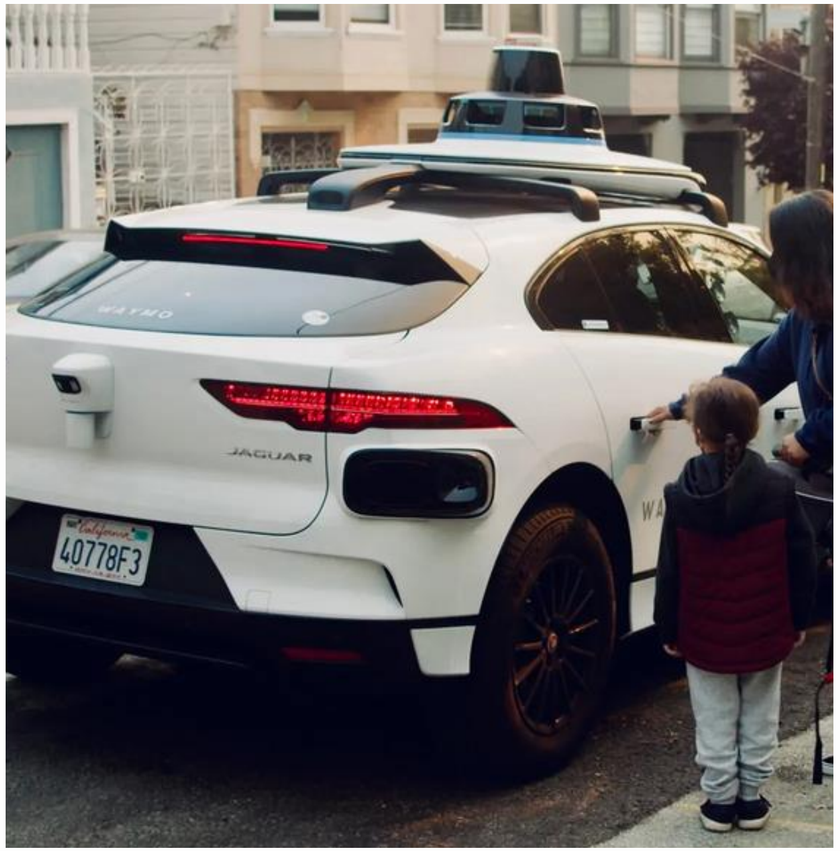
Developed by Tesla (USA)

apollo



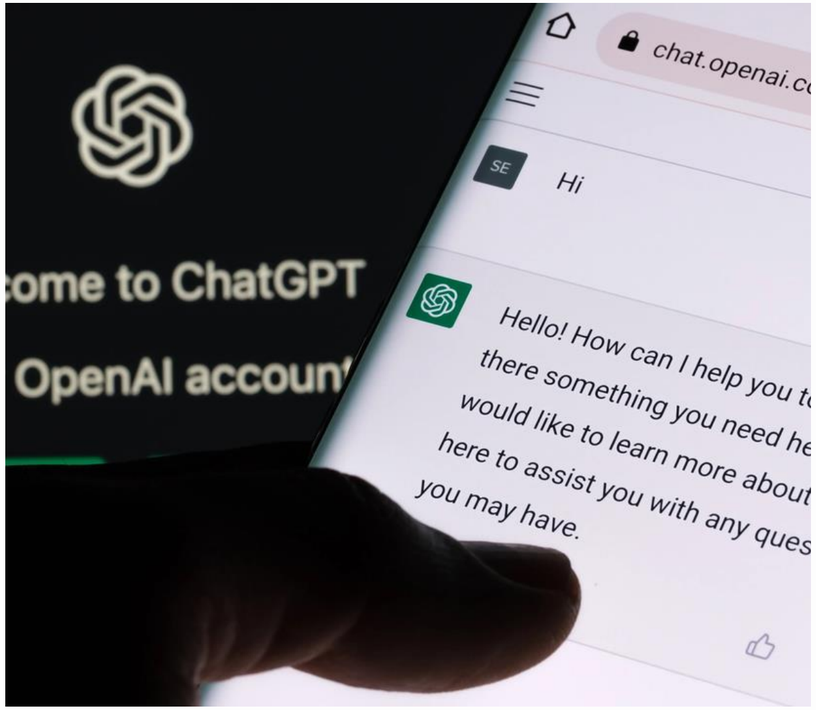

Developed by Baidu (China)

WAYMO




Developed by Google (USA)

# WHAT IS HAPPENING GLOBALLY?
# GPTs




Developed by OpenAI (USA)

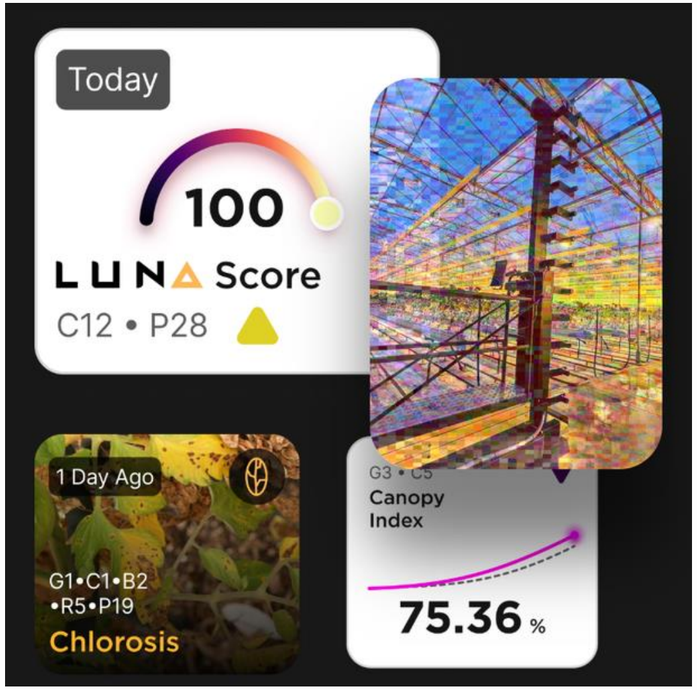
Developed by Google (USA)

Developed by DeepSeek (China)

What Happened to Nvidia?

NVIDIA

# WHAT IS HAPPENING GLOBALLY? AGRICULTURE




**LUNA AI (USA)** optimize crop strategies with automated comprehensive data collection for every plant in the greenhouse.

**Chongho Bridge (China)** provides AI based service in agricultural production, precise planting, traceability of products.

**Small Robot Co (UK)** is revolutionizing farming by leveraging AI, robotics, and machine learning to make farming more sustainable.

# WHAT IS HAPPENING GLOBALLY? EDUCATION

Khanmigo (USA):
On-demand Ai-powered Support For Education. By Khan Academy

Duolingo (USA) Language-learning Application Is A Great Example Of An Ai-based Learning Platform Integrating A Smart Bot To Interact With Students.

松鼠Ai
Squirrel Ai

SquirrelAI (China) provides an AI-based tutoring platform for students.

# WHAT IS HAPPENING GLOBALLY?
# HEALTH

Google Health

Google Health (USA)

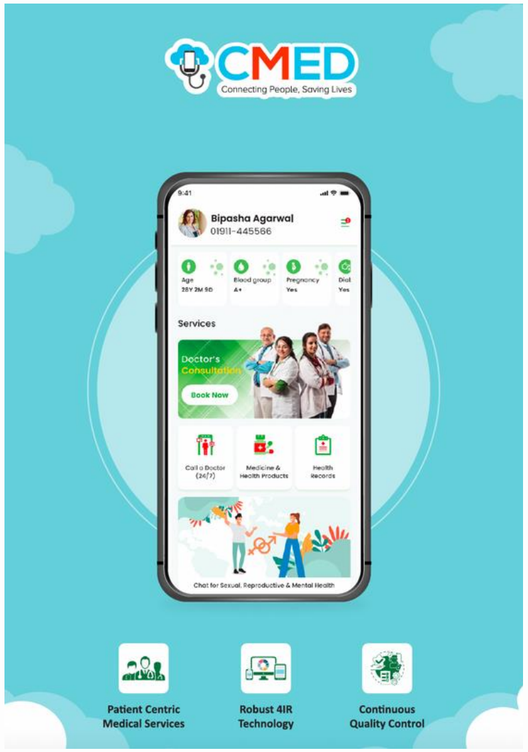
PREDICTING LUNG CANCER

qure.ai

Qure.ai (India)

Early Detection & Management of TB, Lung Cancer, Stroke

SUSASTHO.AI




Susastho.AI (Bangladesh)

AI-driven personalized healthcare platform

## HEALTH Sector is in the Top 3 AI Investment Globally

# WHY AI?

AI solves complex problems faster and smarter.

AI can boost

Productivity

Efficiency

Accuracy

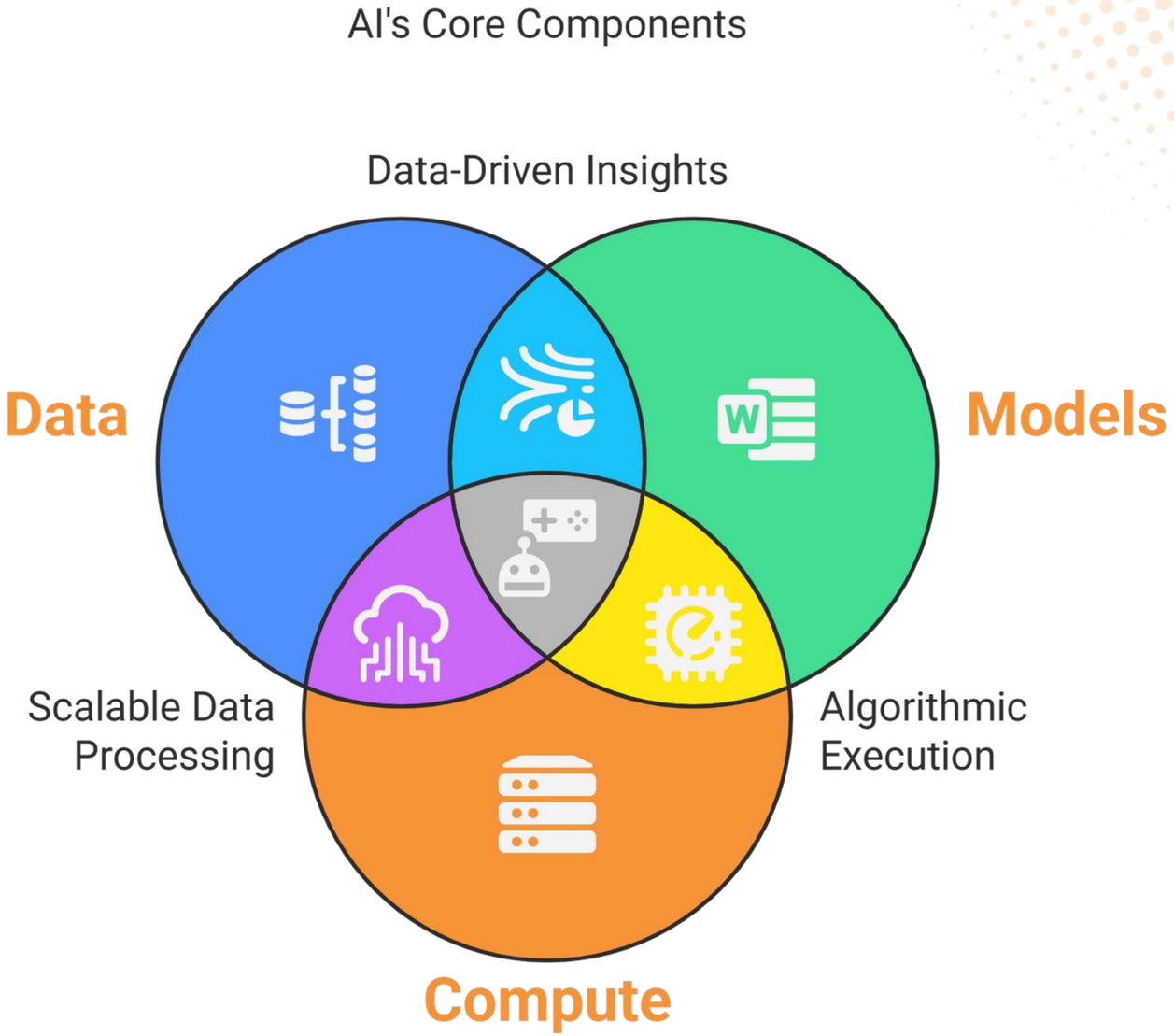
Transparency

AI could contribute up to $15.7 trillion to the global economy in 2030

# What Powers AI?

## AI = Data + Models + Compute

- **Data**
  The raw material — clean, diverse, and context-specific datasets
- **Models**
  Algorithms and foundational AI models that learn from data and solve problems
- **Compute**
  High-performance infrastructure (GPUs, TPUs, cloud) to train and run AI models



# What Powers AI?

## The Five-Layer Cake of AI

### Layer 5: Applications

The end-user products and services that leverage AI to solve problems and provide value.

### Layer 4: Models

The software algorithms and neural networks (including open-source) that form the intelligence of AI.

### Layer 3: Infrastructure

The physical data centers and supercomputers that house, cool, and operate the AI hardware.

### Layer 2: Chips

The specialized semiconductors and processors that perform the complex calculations for AI tasks.

### Layer 1: Energy

The foundational power required to run all AI systems, supercomputers, and data centers.

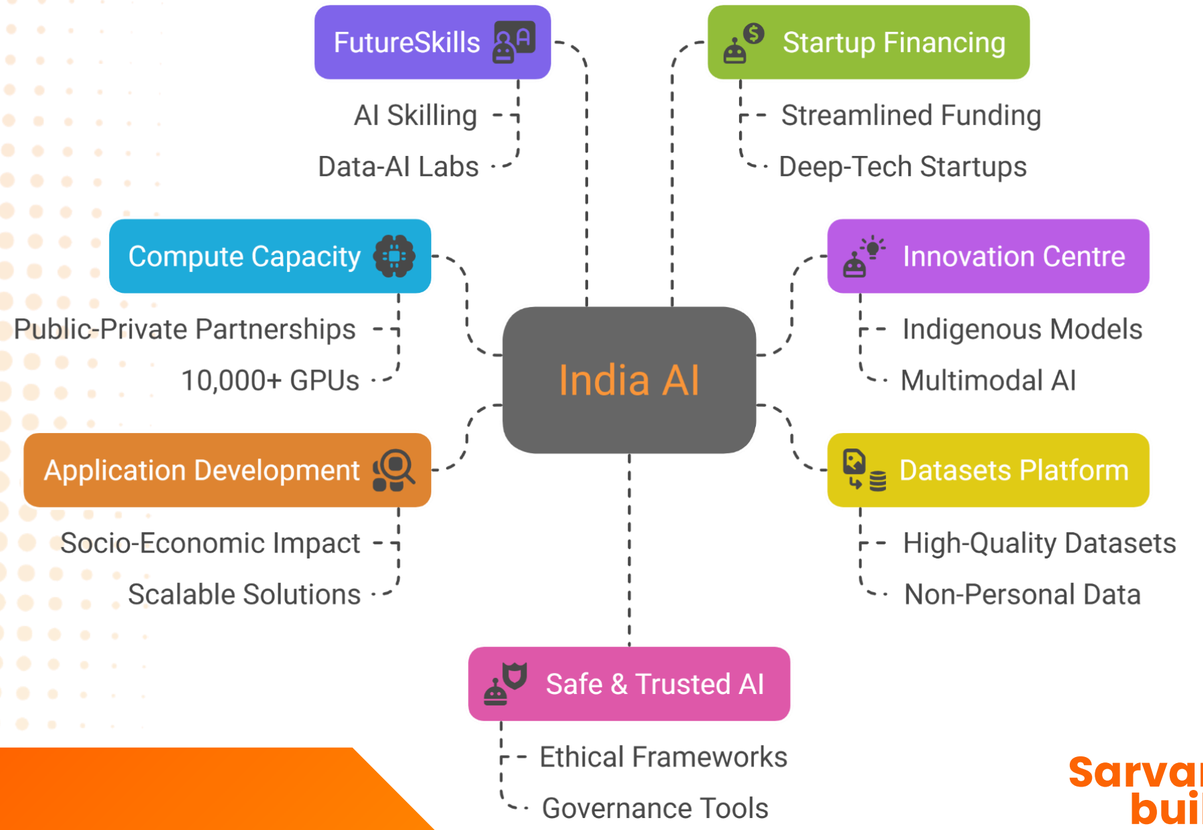
# GLOBAL CASE STUDIES:
## INDIA'S AI MOVEMENT

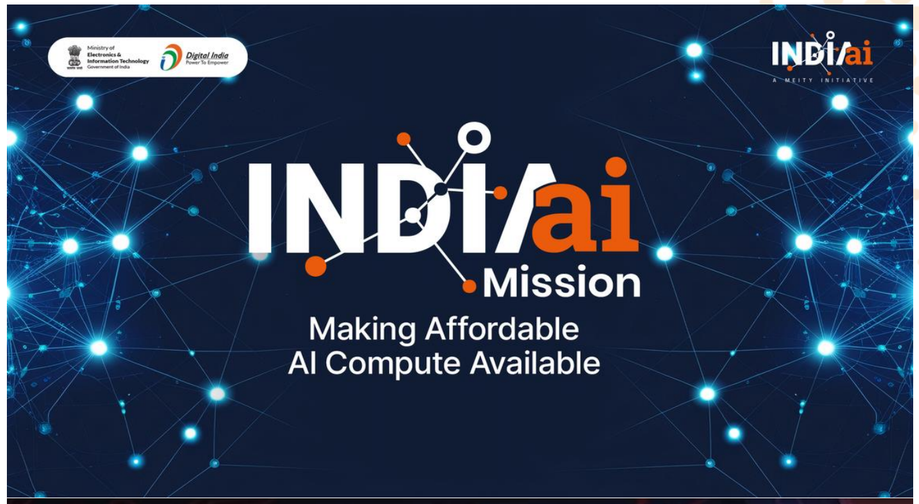
## IndiaAI Mission

## Announced in 2020

**Vision:** Build a robust, responsible, and inclusive AI ecosystem driving innovation, talent, and societal impact.

7 Core Pillers on IndiaAI



**Sarvam AI to receive funding from Indian government to build an indigenous AI large language model (LLM)**

# GLOBAL CASE STUDIES:
## Pakistan's AI Momentum

## National AI Policy 2025

**Announced in 2024 | Approved in 2025**

**Vision:** To build a human-centered, inclusive, and innovation-driven AI ecosystem that empowers citizens, strengthens governance, and enhances national competitiveness.

**1. AI for Governance**
- Make government smarter and more data-driven.
- Deploy AI dashboards for public services.
- Use predictive analytics in planning and decision-making.

**2. Human Capital Development**
- Build an AI-ready workforce.
- Train 1 million AI professionals.
- Integrate AI curricula in schools & universities.
- National e-learning & certification platforms.

**3. Innovation & Startups**
- Nurture local innovation and entrepreneurship.
- Launch National AI Innovation Fund (NAIF).
- Support 1,000 AI startups/products. Incentivize public–private partnerships.

**4. Infrastructure & Compute**
- Strengthen AI research infrastructure.
- Establish National AI Cloud & Data Centers.
- Build regional AI labs across provinces.

**5. Ethical & Responsible AI**
- Ensure trust, fairness, and accountability.
- Create a National AI Ethics Board.
- Develop data privacy and safety frameworks.

**6. AI for Public Good**
- Address societal challenges with AI.
- AI in healthcare, agriculture, climate resilience, and education.

AI ACADEMY
AI FORUM BANGLADESH
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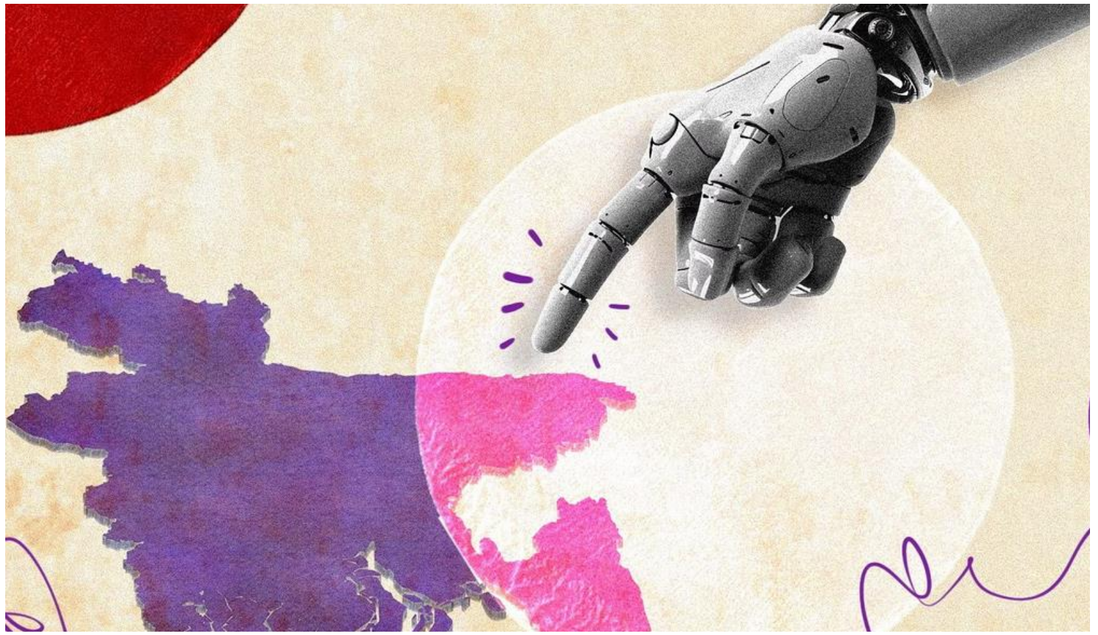
# Albania Appointed the World's First AI Government Minister

# WHERE IS BANGLADESH

- **Still in primary stage.**
- Active in competitions, but **limited large-scale AI development and deployment.**



The Daily Star

# Africa's AI movement and what Bangladesh can learn

Khondaker A. Mamun

Tue May 27, 2025 07:53 PM

For many people, the idea of African nations leading in cutting-edge technology might come as a surprise. The continent has long been

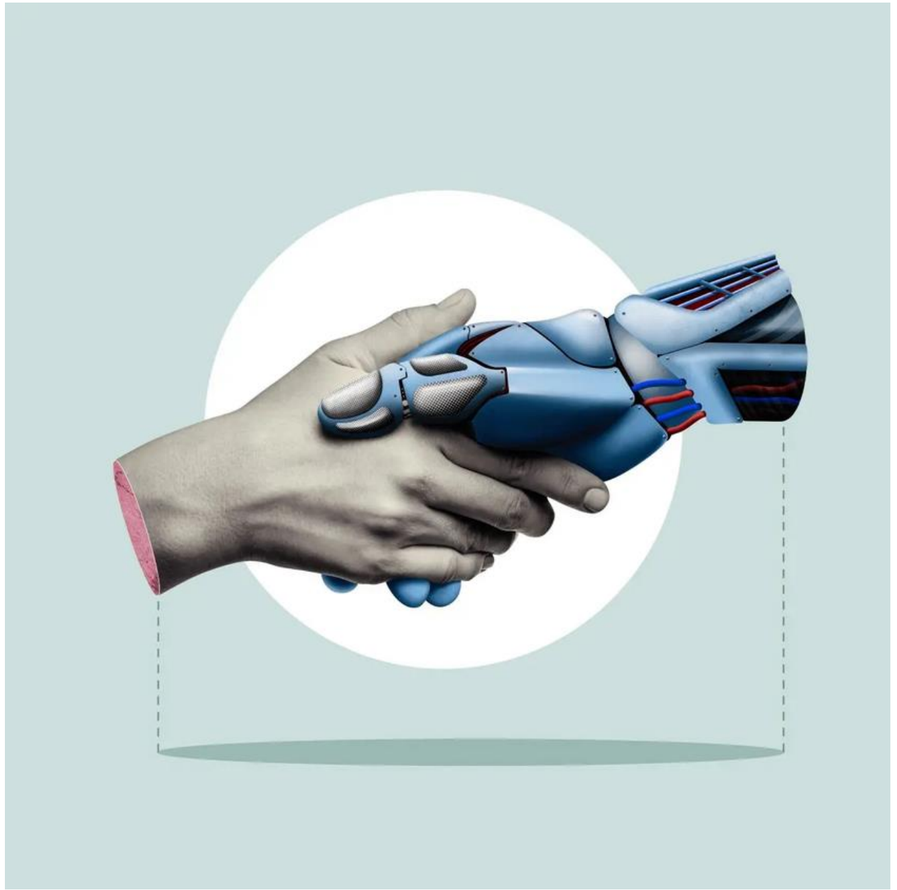
# Challenges for Emerging Economies in AI

- **We have talent** — young, tech-savvy, and globally competitive

- **We have data** — rich, diverse, and locally relevant

**But…**

1. **We lack Infrastructure and Funding** to capture data
2. **We lack compute power** — limited access to GPUs, HPC infrastructure

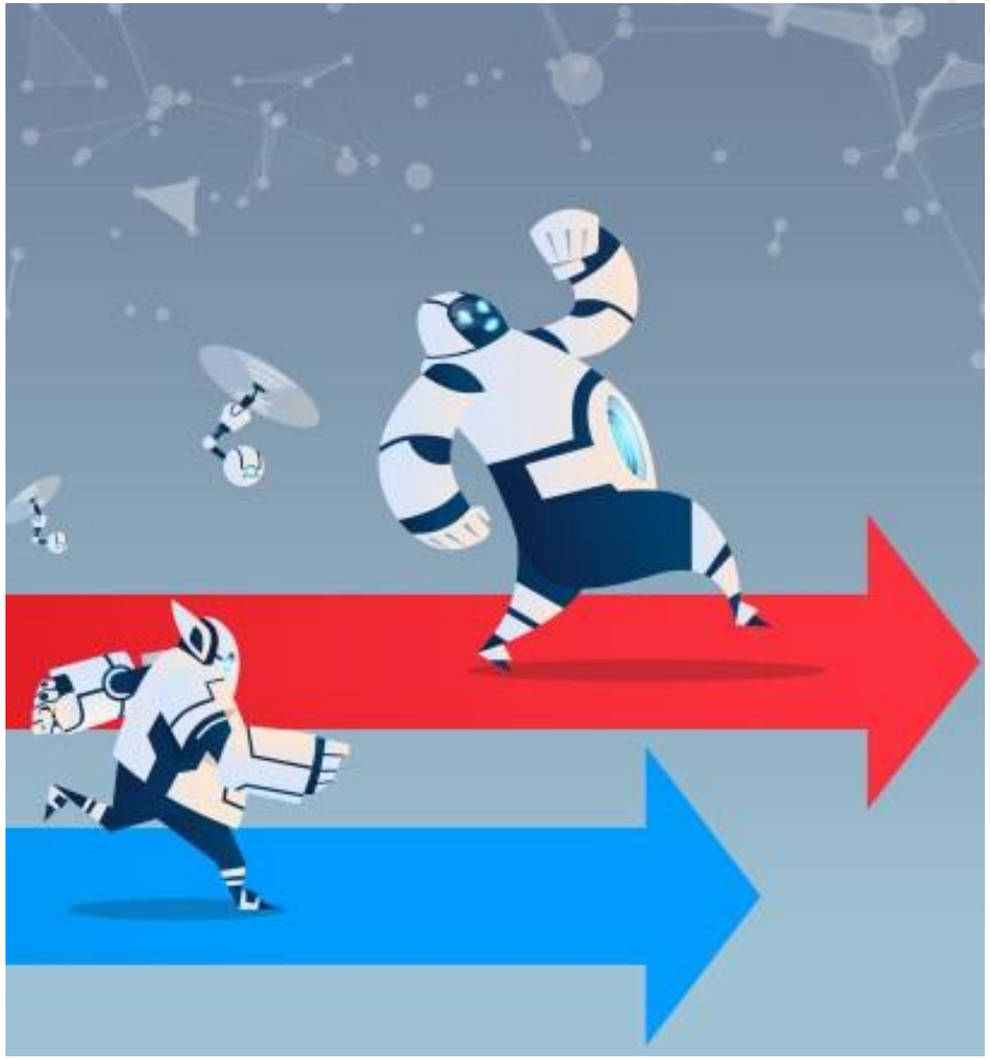


## The risk?

**A growing digital divide** — where AI-rich nations lead, and others follow

# THREATS IF WE DON'T ACT

- Losing competitiveness to the global AI movement.
- Dependency on foreign AI systems, risking digital colonization.



**Without proactive strategy and investment, Bangladesh risks remaining a user of AI, not a creator**

# Do We Want to be User or Creator?

- Be AI consumer only or global AI innovator?
- Use AI models, or build our own?
- Rely on foreign tech — or develop homegrown AI for local problems?

Learn More

**Emerging Economies have the potential to lead in AI for Development.**

প্রথম আলো

সাক্ষাৎকার

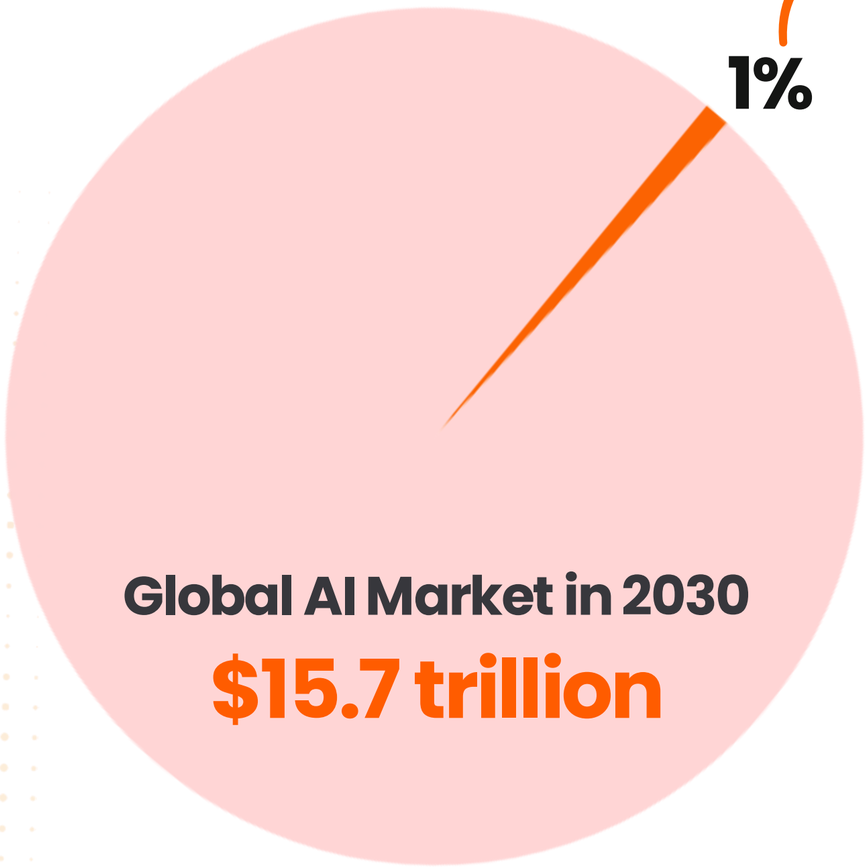
বাংলাদেশ

## এআই প্রযুক্তি গার্মেন্টসের চেয়ে বড় রপ্তানি খাত হতে পারে

কৃত্রিম বুদ্ধিমত্তা (এআই) নিয়ে কয়েক বছর ধরে কাজ করছেন ঢাকার ইউনাইটেড ইন্টারন্যাশনাল ইউনিভার্সিটির (ইউআইইউ) অধ্যাপক খন্দকার আবদুল্লাহ আল মামুন। গবেষণার জন্যই ইউআইইউতে প্রতিষ্ঠা করেছেন অ্যাডভান্সড ইন্টেলিজেন্স মাল্টিডিসিপ্লিনারি সিস্টেমস ল্যাব (এইমস ল্যাব) এবং ইনস্টিটিউট অব রিসার্চ, ইনোভেশন, ইনকিউবেশন অ্যান্ড কমার্শিয়ালাইজেশন (আইআরআইআইসি)। সাক্ষাৎকার নিয়েছেন **মো. সাইফুল্লাহ**

প্রকাশ: ২৪ জানুয়ারি ২০২৫, ১১: ০০

# AI Opportunities for Bangladesh




## $150 Billion Opportunity

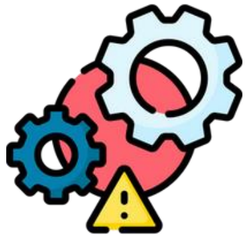
RMG Sector
$42 Billion

AI and Tech Sector
$150 Billion

# THE OPPORTUNITY FOR EMARGING ECONOMIES

## AI can solve:

Corruption



Inefficiency

Public service delivery

Healthcare access

Education

Agriculture

Traffic management

Energy Management

Disaster early warnings

# AI Opportunities for Bangladesh

**AI Policy**

**AI Strategy**

**AI Regulation**

## AI Opportunity Landscapes

### Agriculture & Food Security

AI makes farming smarter with precision, prediction, and better supply chains to boost yields and cut losses.

### Healthcare & Well-being

AI can shift healthcare from reactive to preventive with early screening, personalized risk assessment, and connected digital care.

### Education & Human Resources

AI can personalize learning, upskill teachers, and bridge the industry gap to build a future-ready workforce.

### Public Services and Governance

AI can streamline governance with data-driven decisions, transparent services, and citizen-centric digital access.

### Environment, Climate & Energy Management

AI can strengthen climate resilience with early warnings, smarter risk monitoring, and support for a green energy transition.

RISKS of AI

Responsible and Irresponsible AI Uses

# AI is a Healer, AI is a Killer

**The same technology that saves lives can end them. The difference is how we choose to deploy it.**

## AI as Healer

- Early cancer detection
- Accelerated drug discovery
- Personalized treatment plans
- Mental health support at scale

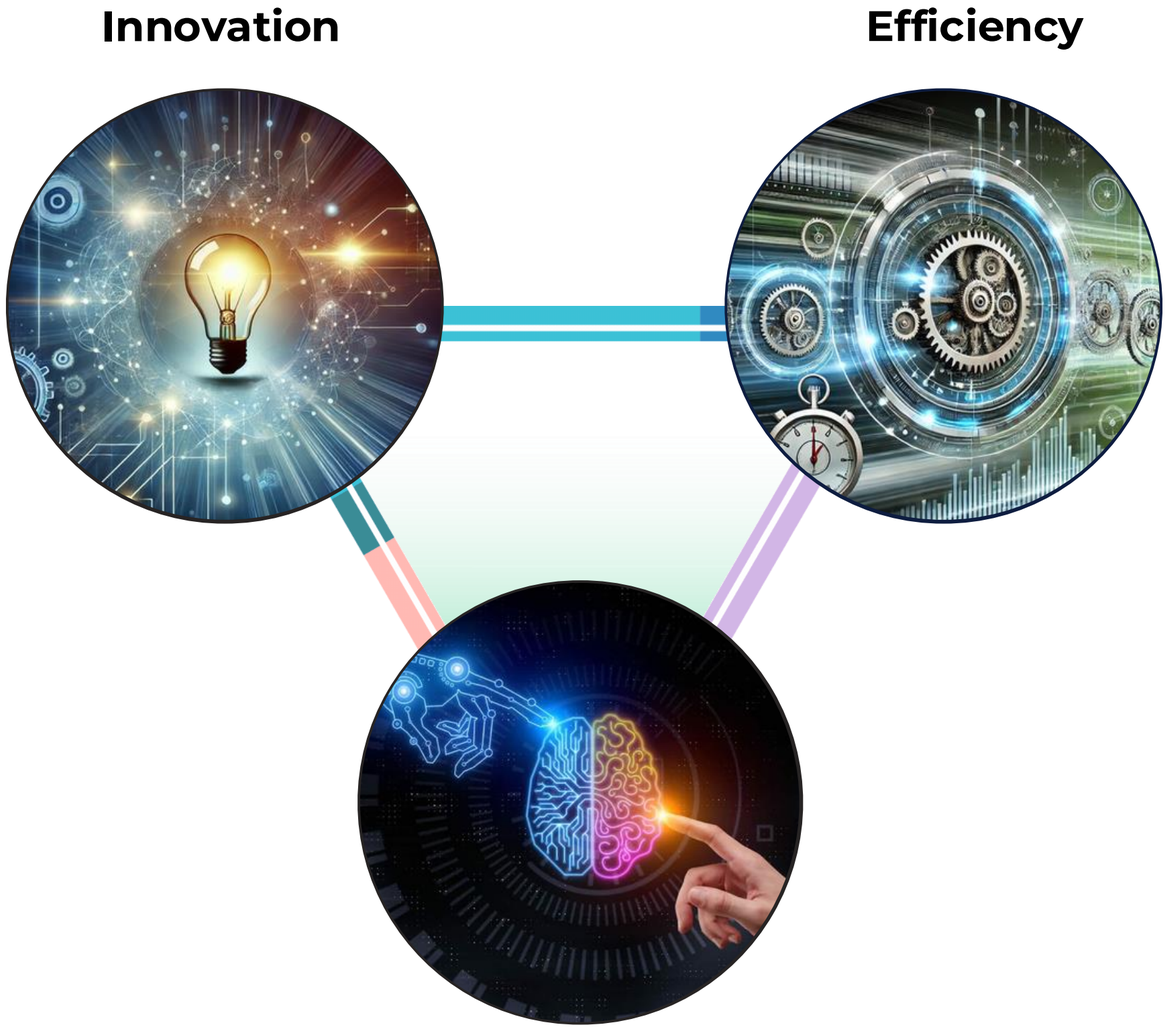
⚠

## AI as Killer

- Autonomous weapons systems
- Algorithmic bias & discrimination
- Deepfake misinformation
- Mass surveillance & control

**The choice is ours.** Responsible AI transformation requires intentional governance, ethical guardrails, and human accountability at every stage.

# The way out


Innovation
Efficiency
Artificial Intelligence
Industry 4.0

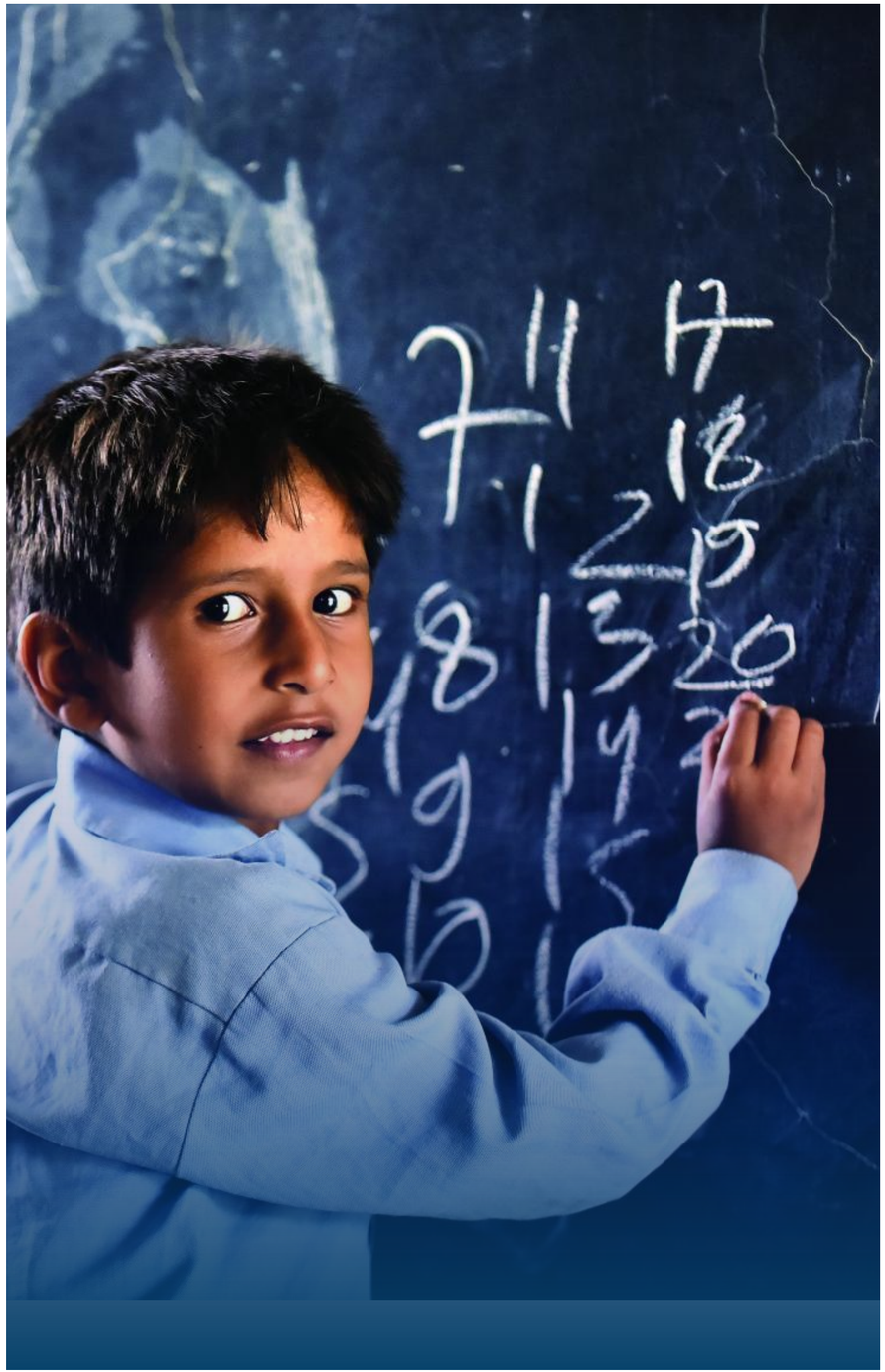
# How AI & 4IR Can Address Challenges

## Expanding Access to Education

## Enhancing Learning Quality and Skills Development

## Boosting Creativity & Employability

## Knowledge Creation & Innovation at Universities



# University as an Innovation Hub

- From degree factory to innovation engine

- Students and Teachers as innovators and knowledge creators

- Research tied to real national problems

- Industry, Government, Academia Collaboration for Innovation Ecosystem

- Ideas move to impact, not shelves

Learning

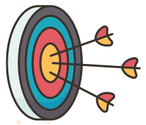
Solving

Building

# Youth as Change Agents

- Youth closest to real problems

- From job-seekers to solution builders

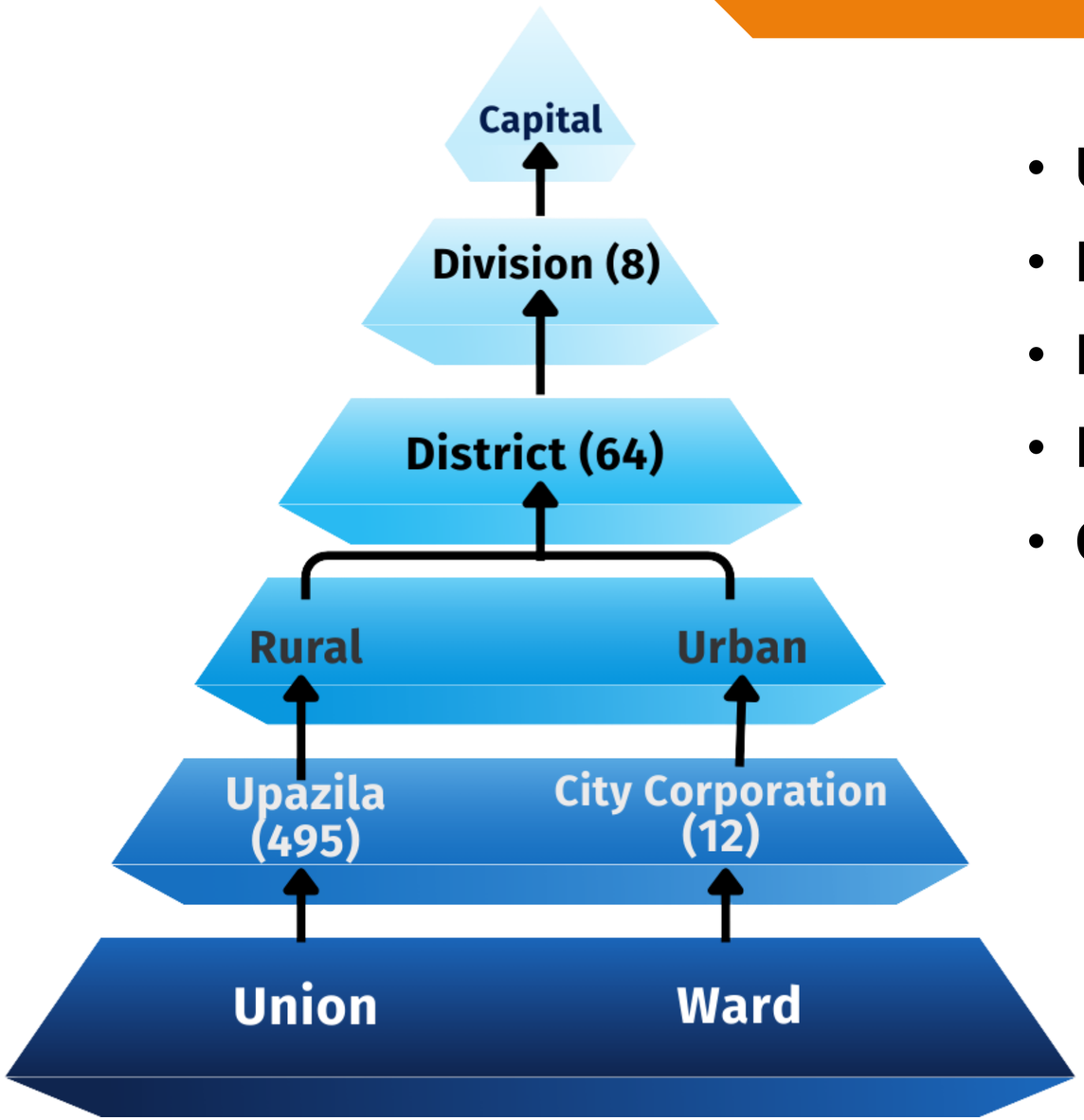


- Local challenges become innovation opportunities

- AI and digital tools multiply youth capacity

- Leadership through problem-solving, not position

# Innovation Transform Union to Capital




- Union as youth-led problem-solving hub
- Local problems solved by local innovators
- Knowledge and solutions flow upward
- Innovation beyond cities and elites
- Capability distributed, not centralized

# Building Equity Through Distributed Power

- **Power moves from central institutions to local communities**
- **Local Community become hubs of innovation and problem-solving**
- **Solutions emerge where problems exist**
- **Inequity falls as communities gain real agency**

**Equity is built by distributing capability, not charity**

# AI for Education

- **Flipped classroom:** Recorded lessons from expert teachers delivered via digital platforms in village schools.
- **Adaptive Tutoring:** AI tutor tracks student progress, identifies weaknesses, and customizes exercises.
- **Assessment & Feedback:** Automated tests provide instant evaluation for students and teachers.
- **Youth Mentors:** Trained local youth guide peers, troubleshoot AI tools, and provide mentorship.

**Impact: Improves learning outcomes, reduces dependence on teacher quality, and builds digital/AI skills.**

# AI for Healthcare

- **AI Health Apps:** Monitor personal and family health, perform risk assessments, and triage remotely.
- **Predictive Diagnostics:** AI identifies high-risk cases and suggests timely referrals.
- **Telemedicine Support:** Connects patients to doctors and specialists without travel.
- **Medical GPT**: Health information and second opinion system
- **Youth Health Workers:** Trained local youth assist with AI tools, home visits, and awareness campaigns.

**Impact: Improves access, preventive care, and survival while building skilled youth workforce.**

# AI for Agriculture

- **AI Advisory Systems:** Mobile apps provide guidance on crop selection, pest control, irrigation schedules, and fertilizer usage based on local soil and weather conditions.
- **Market & Price Intelligence:** AI predicts market demand and connects farmers directly to buyers, reducing middlemen dependency.
- **Resource Optimization:** AI helps plan water usage, crop rotation, and harvest timing for maximum yield.
- **Youth Engagement:** Local youth trained as AI agri-helpers assist farmers, collect data, and run micro-enterprises.

**Impact: Boosts productivity, income, and rural job creation.**

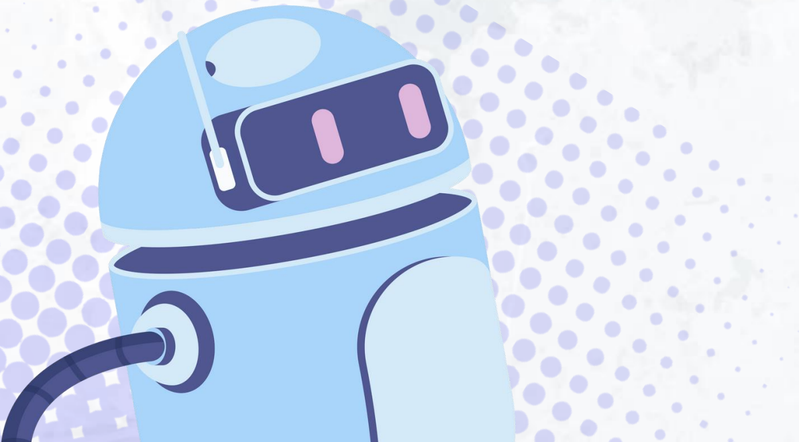
# AI for Climate & Environment

- **Early Warning Systems:** AI predicts floods, cyclones, droughts, and crop risks, sending timely alerts to communities.
- **Resource Management:** AI optimizes water usage, energy, and land resources for sustainability.
- **Environmental Monitoring:** Sensors and AI track soil, water, and weather changes for data-driven decisions.
- **Youth-Led Action:** Local youth implement solutions, run monitoring, and lead awareness campaigns.

**Impact: Reduces disaster impact, increases preparedness, and empowers local innovation.**

# AI for Skills Enhancement & Productivity

# AI Tools for Everyday Work



AI ACADEMY

AI FORUM BANGLADESH

IRIIC
Institute of Research, Innovation, Incubation & Commercialization

UNITED INTERNATIONAL UNIVERSITY

# Easy AI Tools for You

ChatGPT deepseek

Claude Gemini Grok

**AI-powered conversational agents designed to understand queries, generate human-like responses, and assist with tasks across diverse domains**

grammarly QuillBot ProWritingAid

**AI-powered writing assistants that enhance grammar, clarity, tone, and overall writing quality in real time.**

AI ACADEMY AI FORUM BANGLADESH IRIIC UNITED INTERNATIONAL UNIVERSITY

# Easy AI Tools for You

Otter.ai

fireflies.ai

airgram

**AI transcription and meeting assistant tools that automatically record, transcribe, summarize, and organize conversations from meetings, interviews, or calls.**

Canva

Gamma

Napkin

NotebookLM

Pitch

**AI-driven presentation and content design tools that help users create visually compelling slides, documents, and graphics with minimal effort**

# Easy AI Tools for You

CapCut Clipchamp invideo AI Canva

**AI-powered video editing platforms that simplify content creation with video generation, effects, and drag-and-drop tools for creators of all levels**

Notion AFFiNE Reflect

**AI-powered thinking and note-taking tools that help capture, organize, and connect ideas visually for better creativity and insight**

# Easy AI Tools for You

Elicit consensus scite_ SEMANTIC SCHOLAR IRIS.AI

**AI-powered research assistants that help users find, summarize, and evaluate scientific literature and evidence-based answers quickly**

n8n make Huginn Your agents are standing by pipedream Gumloop

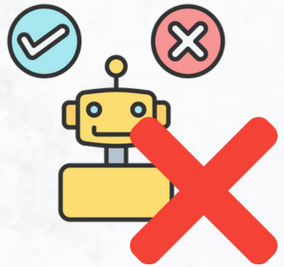
**AI-powered automation tools for workflow automation, API integration, AI agents, or no-code automation and AI agents development.**

# Remember

AI is your assistant, not your boss

AI can support you in reporting, scheduling, predicting, and communication

AI does not replace your leadership, expertise, or final clinical decisions.



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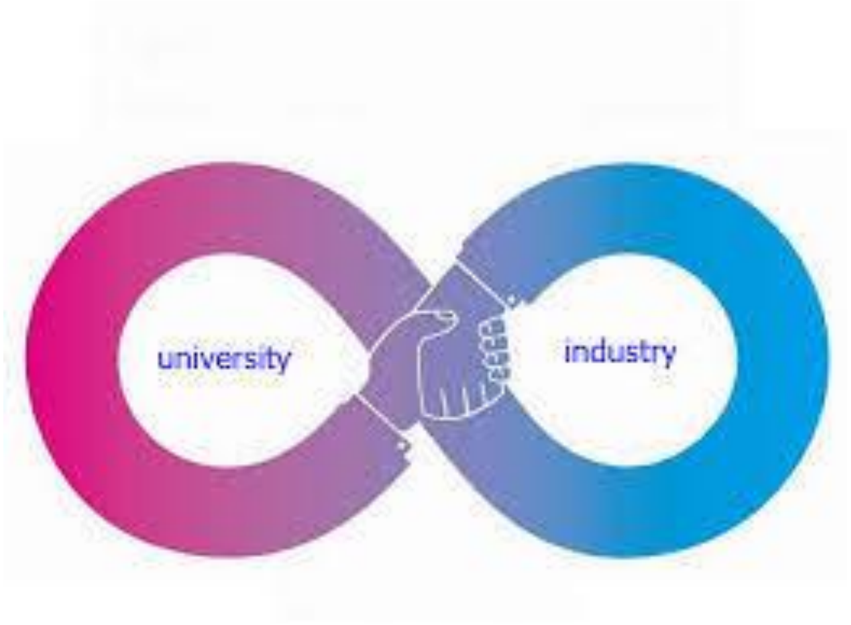
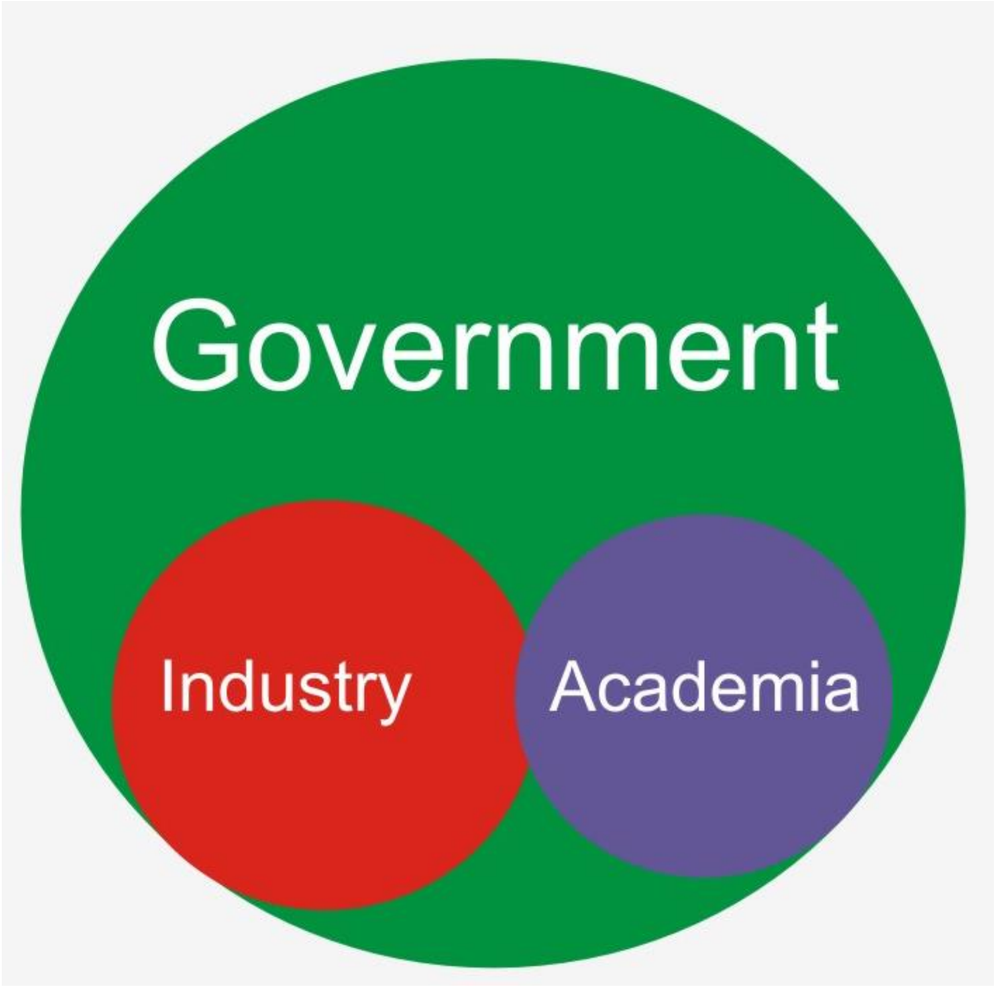
# Living in the AI Era: Embracing the Future

## Where We Live: LMIC

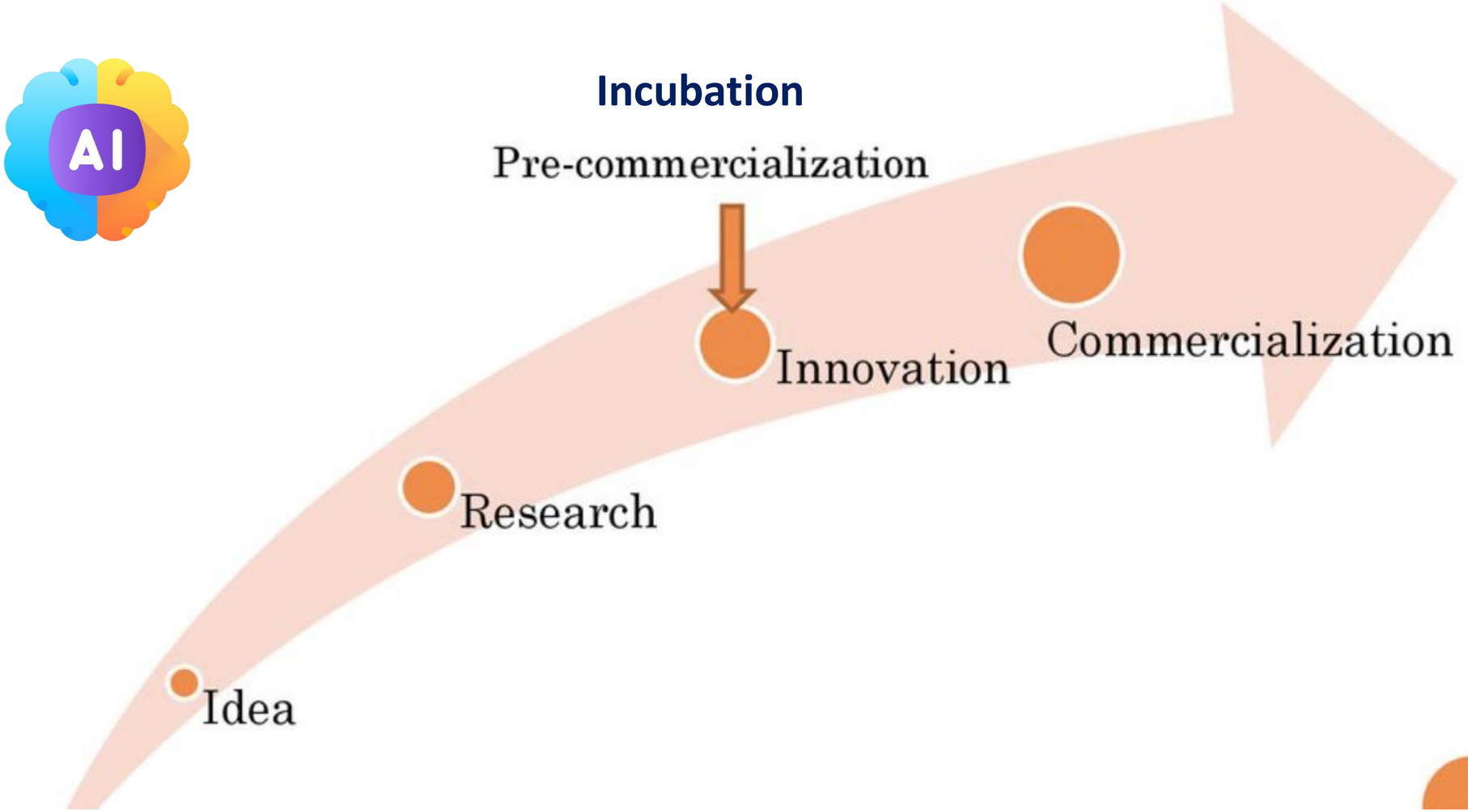



## LMIC : What We Need & Why



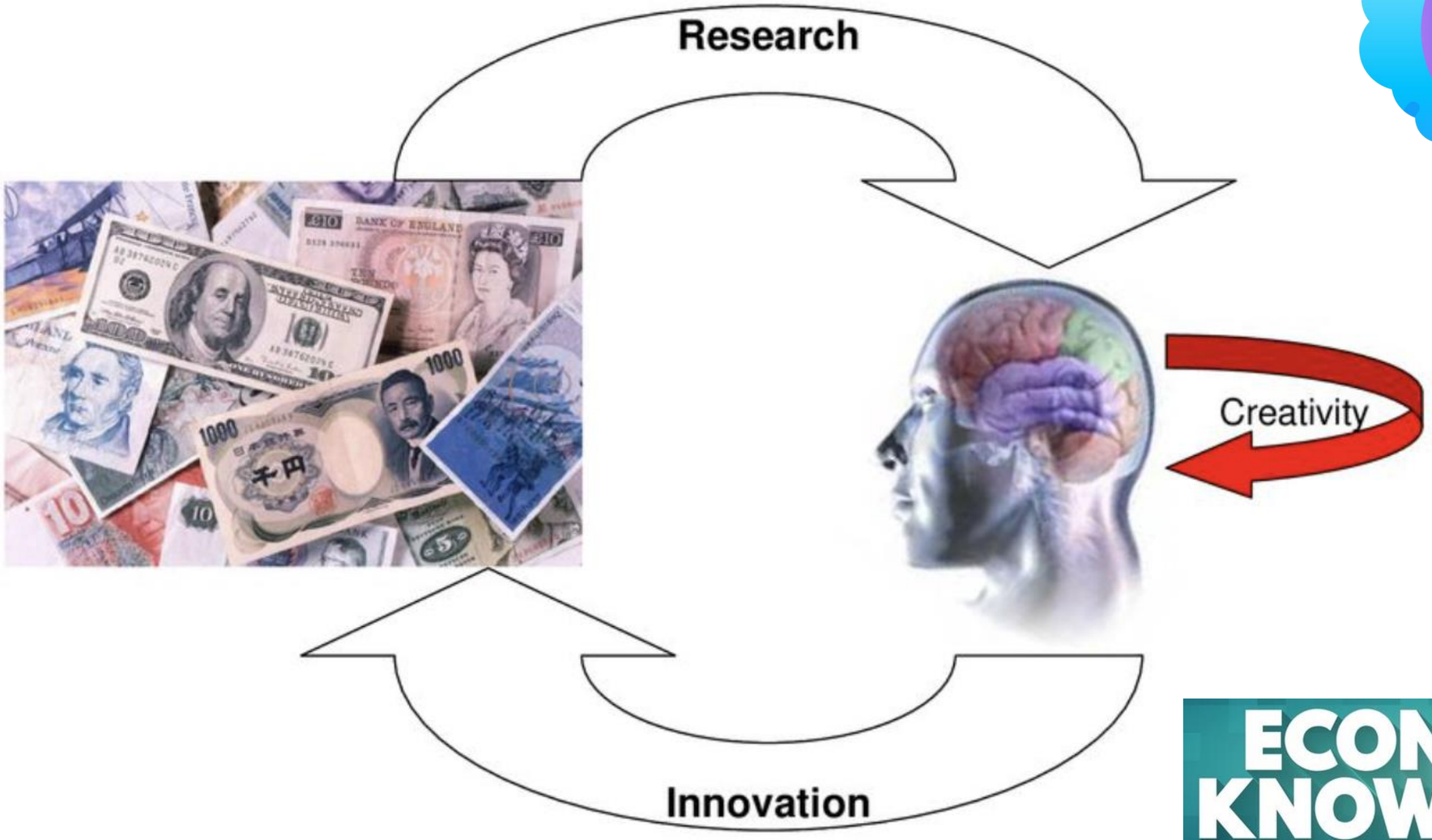

**Fulfilling Individual Need**

# RESEARCH-INNOVATION-COMMERCIALIZATION – GENERAL MODEL




**Research, Innovation, Incubation, Commercialization**

# Research vs Innovation – What is the relationship?




ECONOMY KNOWLEDGE

# Institute of Research, Innovation Incubation & Commercialization (IRIIC)

## A journey from idea to commercialization

2015

3 Research Students

# AIMS Lab

Advanced Intelligent Multidisciplinary Systems Lab

2025

50 Research Engineers/Doctors

3 Million+ USD Grants, 30+ Projects, Trained 200+ Researchers, Impacted 4 M+ Lives

# ব্রেইন কম্পিউটার ইন্টারফেস (বিসিআই) গবেষণাগার

# Brain Computer Interface (BCI) Research Lab

## United International University

বাস্তবায়নেঃ কালিয়াকৈর হাই-টেক পার্ক (এবং অন্যান্য হাই-টেক পার্ক)-এর উন্নয়ন প্রকল্প,
বাংলাদেশ হাই-টেক পার্ক কর্তৃপক্ষ, তথ্য ও যোগাযোগ প্রযুক্তি বিভাগ

# University Innovation Hub (2023)

## Developing Innovation Ecosystem

ICT DIVISION
FUTURE IS HERE

BANGLADESH HI-TECH PARK AUTHORITY

UNITED INTERNATIONAL UNIVERSITY

LAUNCH
PAD
POWERED BY
UIHP@UIU

# HARVARD HSIL HACKATHON 7th Edition

## Building High-Value Health Systems:
Leveraging Artificial Intelligence

HARVARD T.H. CHAN SCHOOL OF PUBLIC HEALTH | HEALTH SYSTEMS INNOVATION LAB

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# HSIL HACKATHON: 7th Edition

# Building High-Value Health Systems: Leveraging AI

**April 10-11, 2026**

Event Hubs in **Boston** (USA), **Buenos Aires** (Argentina), **São Paulo** (Brazil), **Toronto** (Canada), **Santiago de Chile** (Chile), **Beijing** (China), **Bogotá** (Colombia), **Monterrey** (Mexico), **Singapore**, **Istanbul** (Türkiye), **Hanoi** (Vietnam), **Dhaka** (Bangladesh), **Milano** (Italy), **Bandung** (Indonesia), **Accra** (Ghana), **Ogun** (Nigeria), **Delhi** (India), **Cape Town, Johannesburg** (South Africa), **Jeddah, Riyadh** (Saudi Arabia), **Islamabad** (Pakistan), **Sydney** (Australia), **London, Cambridge, Oxford** (UK), **Toronto** (Canada), **Milan** (Italy), **Kuala Lumpur** (Malaysia), **Kigali** (Rwanda), **Berlin** (Germany), **Cairo** (Egypt), **Harare** (Zimbabwe),.....

**Organized by:**

HARVARD T.H. CHAN SCHOOL OF PUBLIC HEALTH
Health Systems Innovation Lab

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# Technologies

AI & Data Science

Brain-Computer Interface (BCI)

Robotics & Automation

Augmented Reality (AR) Virtual Reality (VR)

# Applications

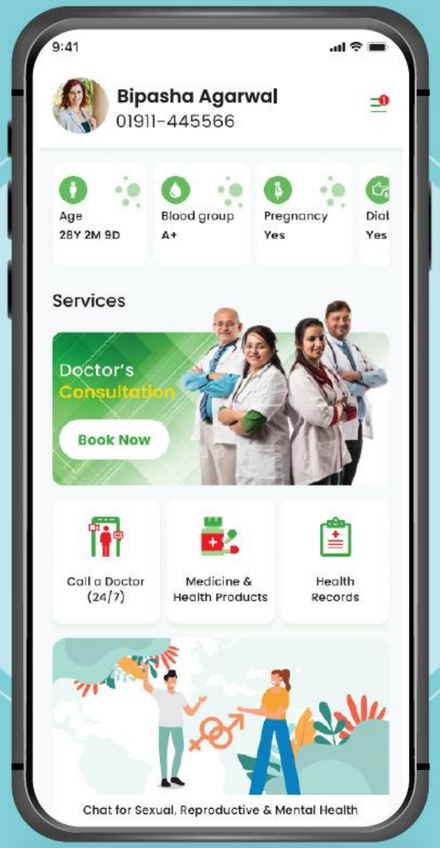
Health & Wellbeing

Education & Skills Development

Energy, Industry & Business

Agriculture & Climate

CMED
Connecting People, Saving Lives




Patient Centric Medical Services

Robust 4IR Technology

Continuous Quality Control

SUSASTHO

# Intelligent General Practitioner (iGP), an AI-powered, HWs Driven, People-Centered Model to Revolutionize Primary Healthcare Delivery in Bangladesh and LMICs for UHC

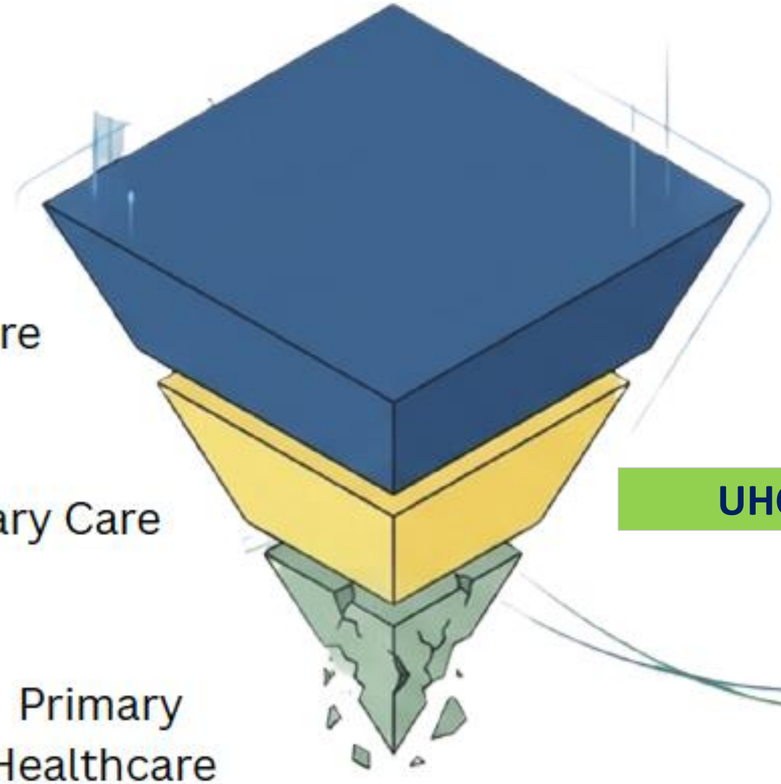
INTELLECTUAL PROPERTY
PATENTED
INTELLECTUAL PROPERTY

https://youtu.be/wAohCAfybio?si=7EaX9xDtupwyhaav

# Re-Architecting Healthcare: The Flipped Pyramid

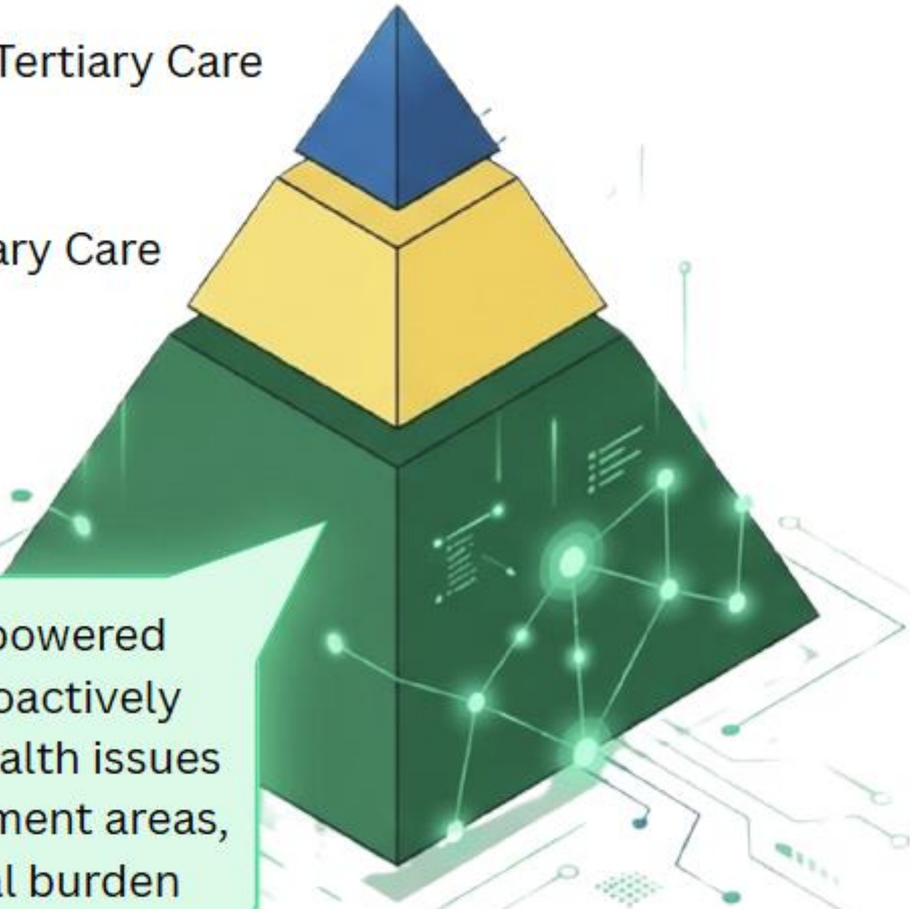
***PHC Based UHC : The Framework***

Current Senarion in provision of health services in Bangladesh




**UHC**

In proposed iGP framework




iGP enables AI-powered ecosystem to proactively manage 90% of health issues with in local catchment areas, reducing hospital burden

***Similar to NHS UK***

SUSASTHO.AI
CMED
Connecting People, Saving Lives
WHITE ANGLE HOUSE
BUSINESS CENTER
MERU
NYERI
NAKURU DIRECT
KARATINA DIRECT
Watch Here: https://youtu.be/p9q4nEQlQhk?si=8mULZjMf3ceZjZ6P

# AI Innovation for Accessible, affordable & Quality Healthcare

Doorstep Service

Personalized Care

Clinically validated Answer of Health Queries

Teleconsultation (Audio/Video)

## AI Engine

**Personalized Health Account**

Monitoring system to ensure personalized health record (EHR) and follow up

**Risk Assessment**

Clinical Decision Support System (CDSS) & Machine learning (ML) for evaluating risks

**Triage & Referral Systems**

Real-time triage & referral utilizing ML and Large Language Model (LLM)

**GenAI for Health Education**

LLM based model for health education, awareness & optimal healthcare guidance

## Inclusive Design

**Multilingual Capability**

Supports Bangla, Banglish, and English

**Clinically Validated Dataset and AI Engine**

Knowledge base for health education, disease symptoms and risk assessment

**Low Bandwidth Optimization**

Designed for both urban and rural areas (online & offline)

BILL & MELINDA GATES foundation

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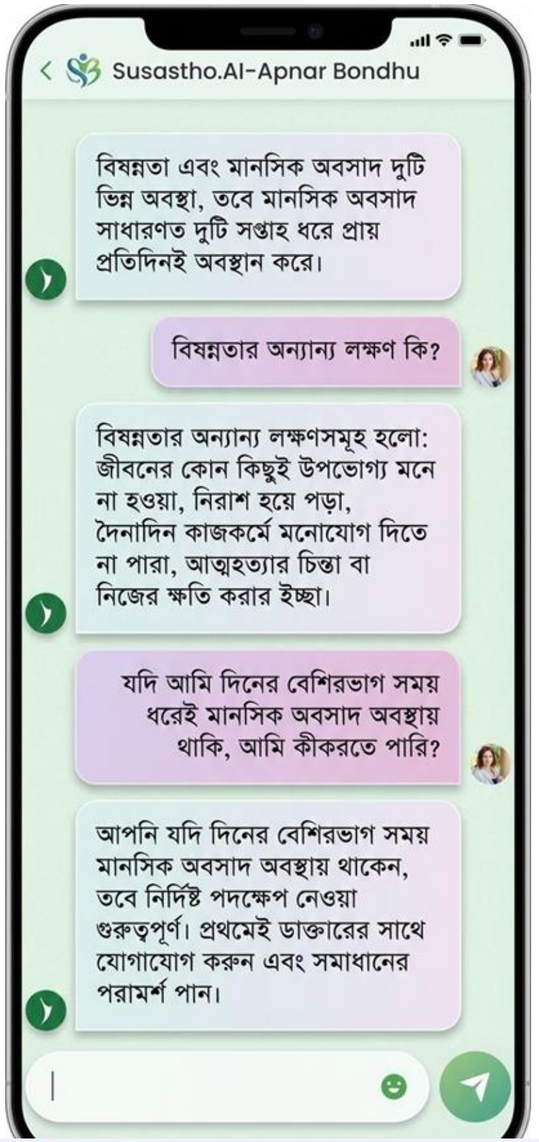
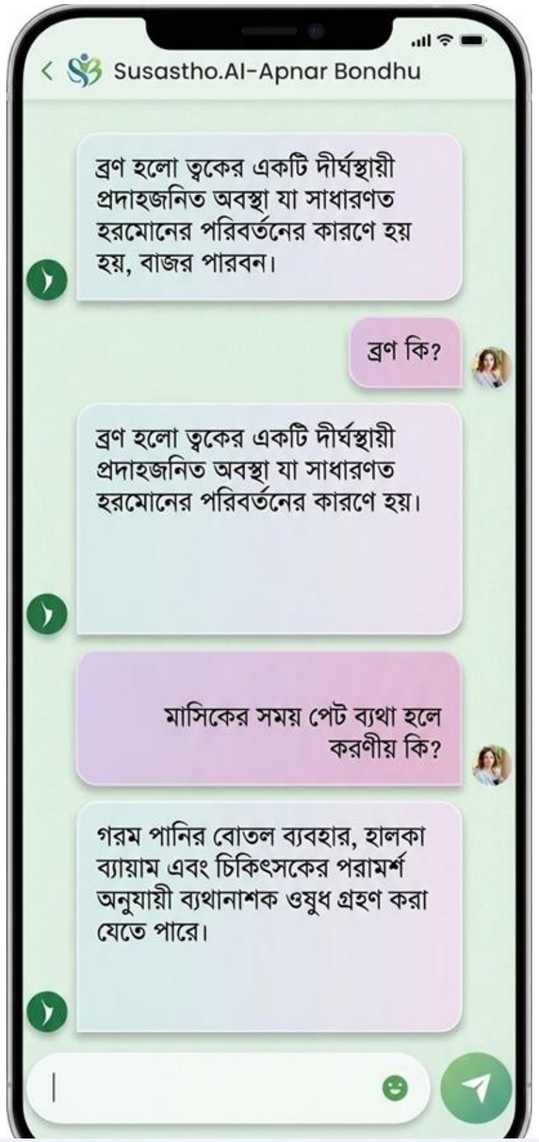
# Susastho.ai usage examples

Education & Screening

Risk Assessment & Referral

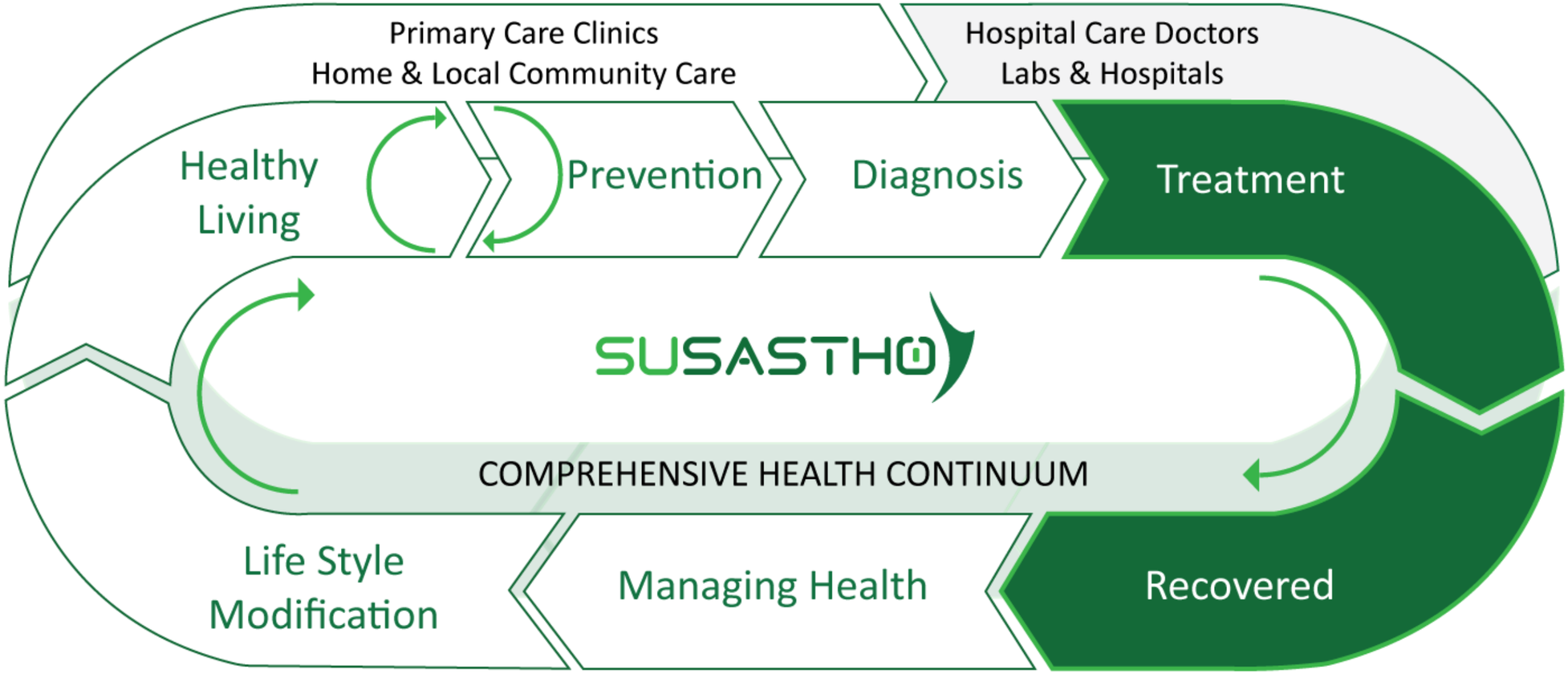
Text & Voice Enabled Chat



# AI Innovation for Accessible, affordable & Quality Healthcare

## Theory of Change




**Periodic Health Check/ Homecare/ Remote Monitoring**
**Risk Assessment & Healthy Lifestyles Advices**

## Continuity of Care through AI

# AI Innovation for Accessible, affordable & Quality Healthcare

**2.5M+** Rural Populace Served

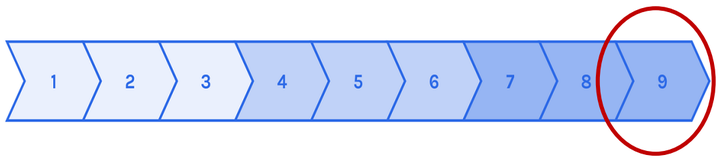
**0.58M+** Urban Populace Served

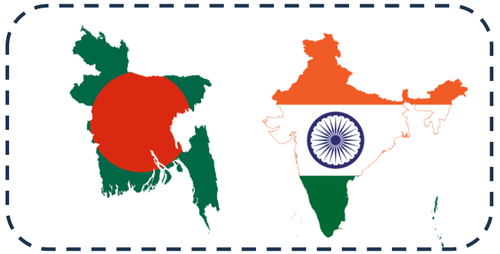
**4 M+** Digital Health Accounts

**3000+** Female Health Workers

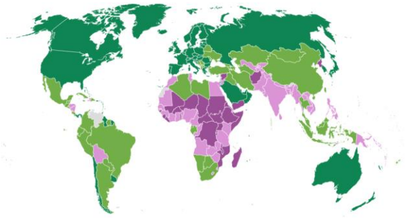
**99.5%** Service Satisfaction Rating

Technology Readiness Levels



Operational in LMIC Settings



Scalable across Globally

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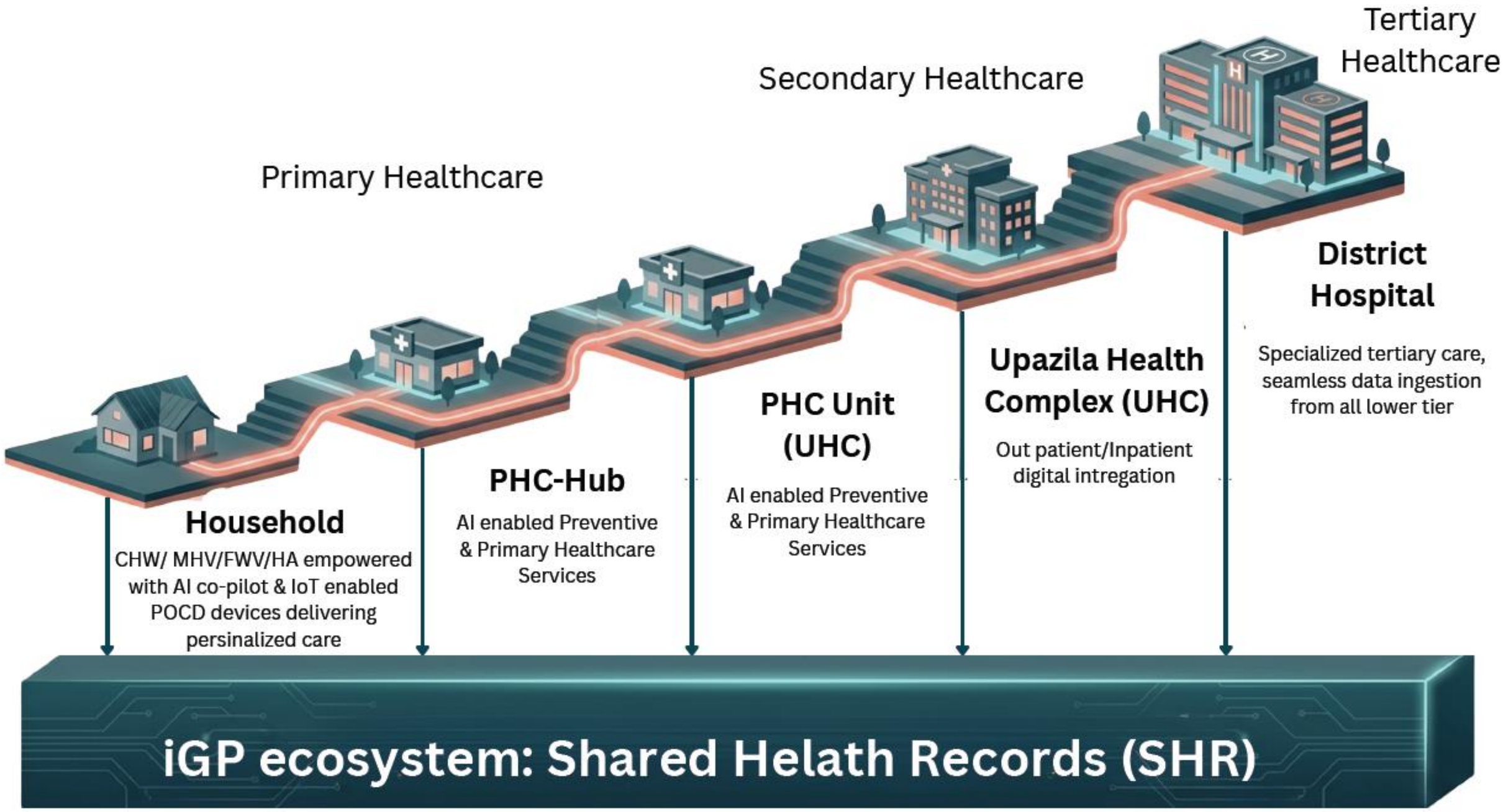
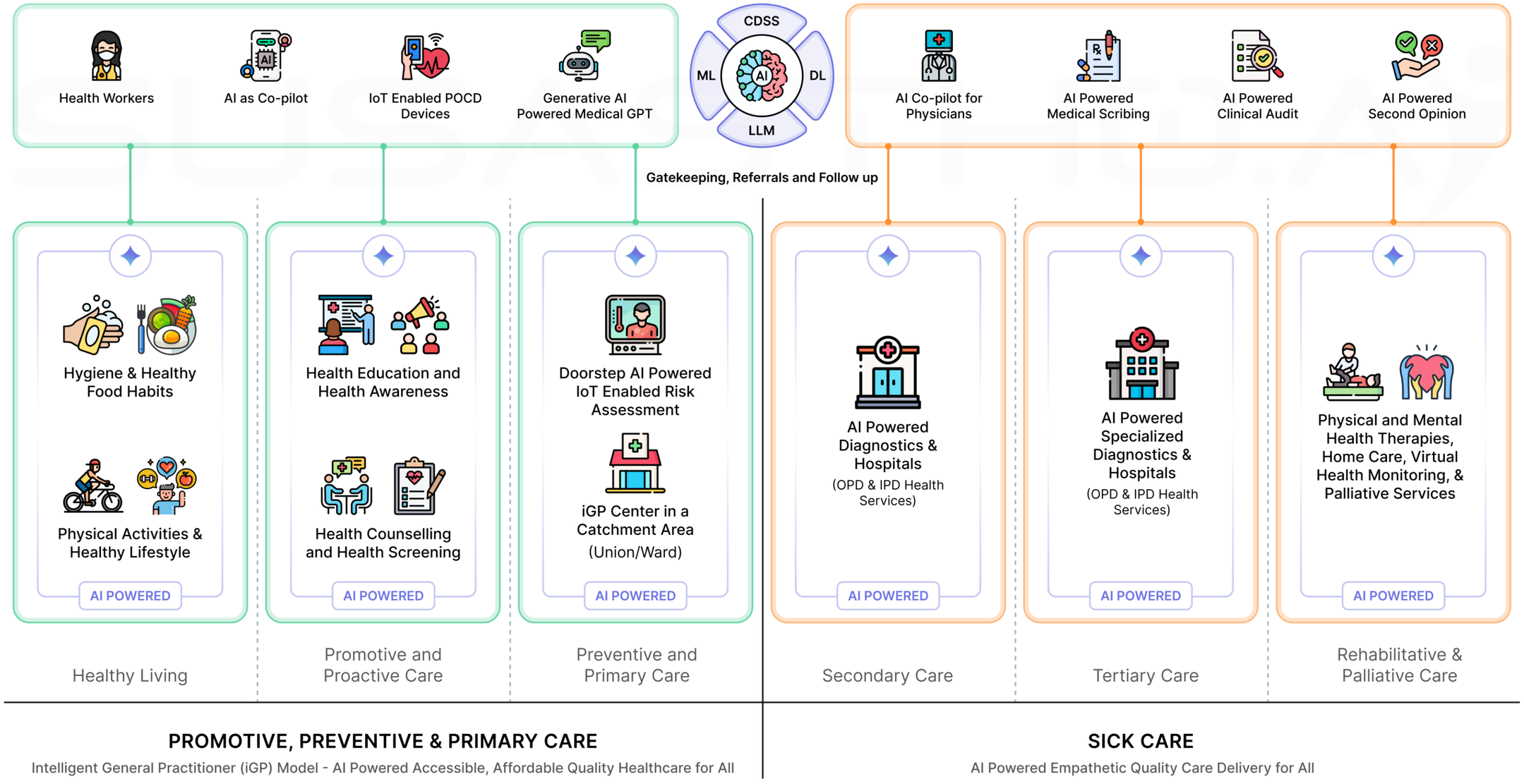
# AI Innovation for National Healthcare Ecosystem Building



# AI Innovation for Accessible, affordable & Quality Healthcare

## Human-Centered AI: Defining the Role of Human-in-the-Loop

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# PrescriptionGPT: Reimagining Our Personalized Healthcare Solutions with AI

## The Crisis: Systemic Healthcare Gaps

### The Illegibility Barrier

Poorly handwritten prescriptions mixing local and medical terms lead to frequent misinterpretation.

### High Medication Non-Adherence

Patients often self-adjust doses or stop treatment prematurely once initial symptoms improve.

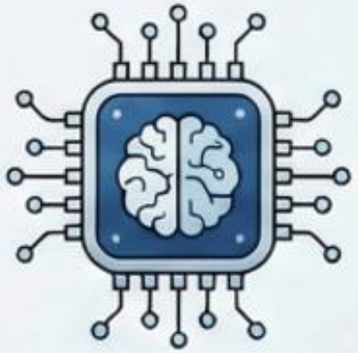
### Dangerous Drug Interactions (DDI)

A lack of point-of-care screening fails to prevent toxic combinations of drugs and food.

## THE PrescriptionGPT AI SOLUTION

### 95% OCR Accuracy Target

Advanced AI recognizes both printed and handwritton prescriptions in mixed Bengali-English formats.



| | |
|---|---|
| Image Capture to OCR: | < 2 Seconds |
| Full Prescription Analysis: | < 10 Seconds |
| Complete Report Generation: | < 15 Seconds |

**PrescriptionGPT Platform**

### Real-Time Interaction Checking

An automated engine cross-checks drugs against DGDA databases for dangerous interactions and allergies.

### 24/7 Bilingual AI Counseling

A culturally adapted conversational assistant provides dosage, diet guidance, and health literacy support.

## Projected Impacts

**40%**

**Reduction in Errors**

Target decrease in medication errors through automated safety checkers and personalized counseling.

**$500M**

**Annual Savings**

Potential household savings achieved by identifying and substituting affordable generic medications.

**1 Million**

**Target Users**

Planned user base within the first year of rollout in the Bangladesh market.

Activate Windo

# SuSastho.AI: Bridging the Adolescent Health Gap
# A Voice-Enabled, Evidence-Based Medical Copilot for Bangladesh

## Problem & Motivation

**63%**
of Bangladeshi adolescents lack SRH services

**Cultural Stigma:** Prevents open discussion of sexual and mental health.

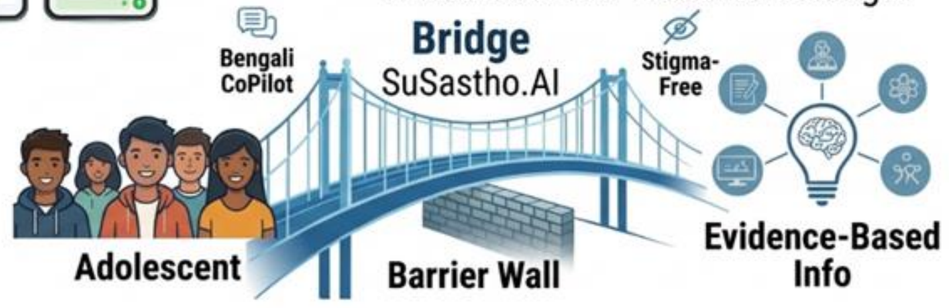
**Resource Gap:** Only 13% receive any form of mental health support.

### Barrier Wall

## The Solution

### Key Features

- **Stigma-Free:** Private, judgment free interaction for sensitive topics.
- **Accessible:** Supports both text and speech (Bengali) for low-literacy inclusivity.
- **Scalable:** 150x lower cost than GPT-4, making it viable for low-resource settings.




## The Technology

**Llama 3**
**The Core:** Fine-tuned Llama 3 model using Chain-of-Thought (CoT) reasoning.

**The Knowledge:** RAG pipeline fueled by 2,000+ clinically validated Bengali documents.

**The Interface (Input):** Speech-to-Text: Wav2Vec.

**The Interface (Output):** Text-to-Speech: VITS.

**86.7%**
**The Result:** 86.7% Overall Accuracy as verified by clinicians.

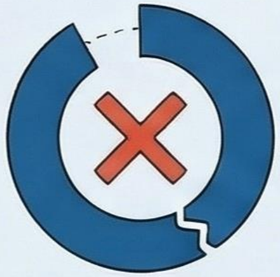
## Impact & Future

**Current Success:** 16.1% increase in response correctness over baseline models.

**Future Vision:** Expanding to newborn/elderly care and nationwide triage systems.

# Culturally and Linguistically Adaptive Conversational AI for Mental Health Screening and Triage in Bangladesh

## THE PROBLEM: AN ENTRY-TO-CARE CRISIS



**92%**
**Treatment Gap**

In Bangladesh, about 1 in 5 adults
&
1 in 8 children live with mental health conditions.

### The Barrier of Stigma and Access

Cultural stigme, fear of jodgment, and the concentration of providers in major cities prevent early help-seaking and symptom recognition.

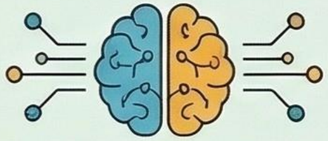
### A Massive Workforce Shortage

The ratio of psychiatrists to patients is negligible, making traditional one-on-one clinical screening impossible to scale for the whole population.

## THE SOLUTION: A SCALABLE ENTRY PATHWAY

### AI-Powered Conversational Triage

A scalable, clinically governed chatbot that serves as the first point of contact for mental health screening and guidance.



### Risk Stratification

The system analyzes user conversations to assign a risk tier (Low, Medium, or High) and recommends specific action pathways.

**Low Risk** (Self-Help Resources)

**Medium Risk** (Digital Therapeutics/ Counseling)

**High Risk** (Urgent Referral to Clinician)

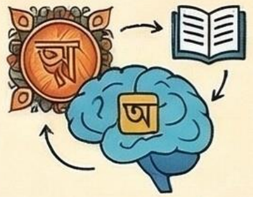
### Human-in-the-Loop (HITL) Safety

Licensed psychiotrists audit 100% of system interactions to ensure all AI generated advice remains clinically accurate and safe.

## THE TECHNOLOGY: CULTURALLY ADAPTIVE AI

### Multilingual & Code-Mixed NLP

The AI is designed to understand Bangla, English, and "code-mixed" messages (switching between languages), reflecting how people actually speak in Bangladesh

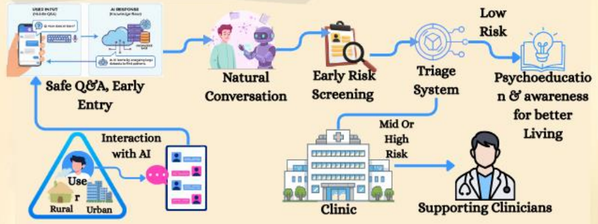


### Cultural Knowledge Integration

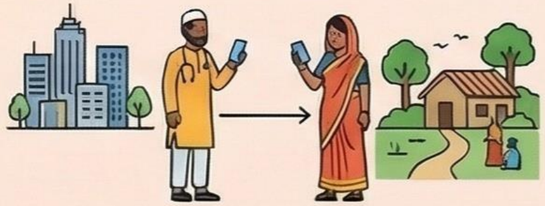
Uses a specialized "Cultural Knowledge Graph" to provide advice that respects local idioms and metaphors of distress rather than relying on generic Western-centric content

### Offline-First Mobile Design

Engineered to function in areas with unstable internet, ensuring rural populations can access the tool via standard mobile devices.




## THE IMPACT: TRANSFORMING HEALTHCARE DELIVERY

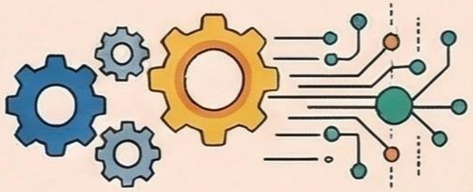


### Health Impact: Reaching the Underserved

The system extends services beyond major cities to rural areas and reduces pressure on overbordened clinicians by automating the initial screening process.

### Social Impact: Breaking the Silence

Provides a private, anonymous platform that encourages early engagement with care, promotes mental health awareness, and actively reduces social atigma.



### Economic Impact: Boosting Productivity

By enabling early intervention, the tool helps reduce the long-term economic burden and productivity fuss caused by untreated mental health disorders.

## THE CRISIS

**9,000**
Deaths/year from medication errors

**4**
Errors per prescription in Bangladesh

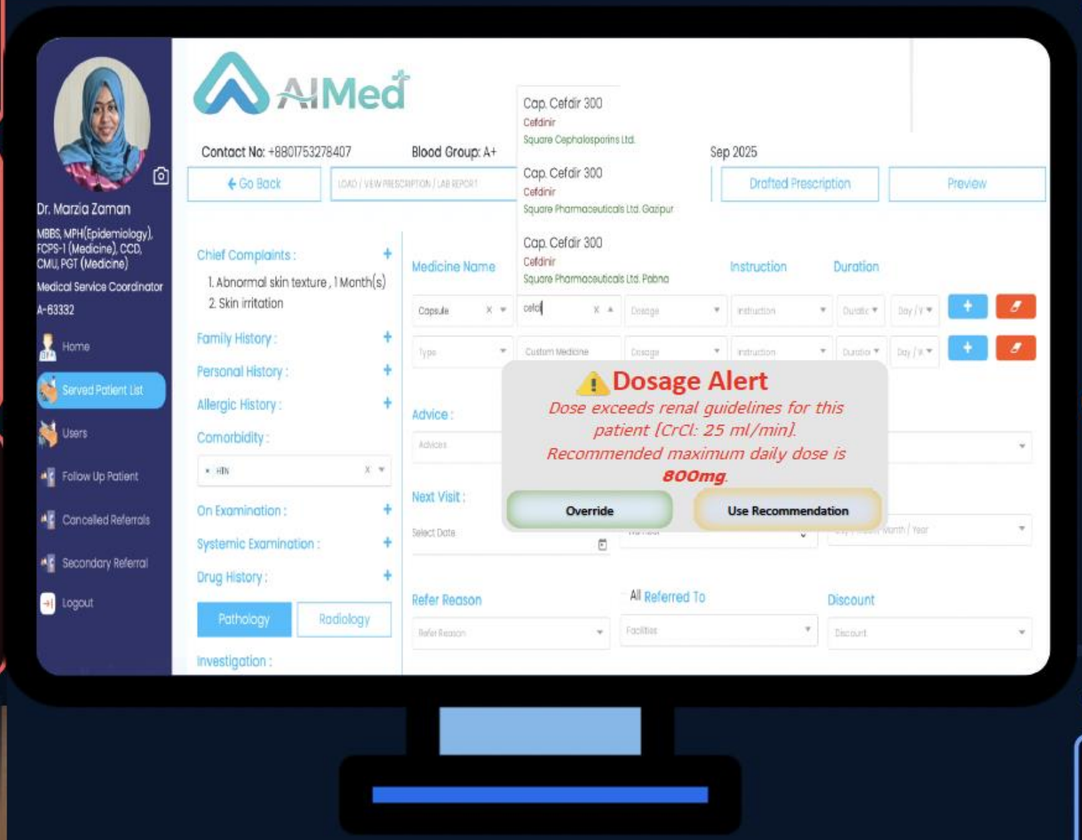
**21%**
Errors arise during prescribing stage

**48s**
Avg doctor time per patient in Bangladesh

**Siloed Data · Alert Fatigue · Time Pressure**
It's not bad doctors. It's overloaded systems — and patients pay the price.

# Intelligent Prescription Co-Pilot

EMPOWERING DOCTORS · PROTECTING PATIENTS




## HOW IT WORKS — The 4-Step Safety Flow

**01** Doctor inputs Prescription Data →

**02** AI Activates
WHO, FDA & medical literature checked →

**03** Risks Flagged →

**04** Suggest Corrections
Doctor approves

## OUR SOLUTION

### An AI engine embedded inside hospital prescribing software

We don't ask doctors to change their workflow. AIMed integrates directly into ERP systems they already use — becoming a silent guardian that catches errors before they reach the patient. Not a standalone tool. A feature inside the software doctors already trust.

SAFETY CAPABILITIES

- Disease-Specific Rx
- Drug–Drug Interaction Alert
- Dose Verification
- Side Effect Warning
- Error Detection
- Renal Dose Adjustment
- Real-time · Zero Latency
- Error Detection

## ★ THE IMPACT

**2.5M+**
Prescriptions safeguarded/yr

**Zero**
Workflow disruption

**50%**
Fewer Rx errors

**Every**
Patient protected

# AIMScribe: AI-Driven Scribing System with Prescription Counseling Tools to Streamline Clinical Documentation, Enhance Medication Safety, and Patient Empowerment in Bangladesh

## THE PROBLEM

**1:1600**
Doctor-to-Patient Ratio

**40%**
of Clinical Hours Lost to Paperwork

**63%**
Physician Burnout Rate

Documentation burdens and constant time pressure have caused a majority of healthcare providers to report severe burnout symptoms.

## THE SOLUTION

**Standardized Prescribing Framework**

**Unified EHR Integration**

Synchronizing medical data into cloud-based EHR for seamless access.

**Automated Prescription Generation**

Generative AI for structured digital prescriptions, eliminating handwriting risks.

**AI-Assisted Medical Knowledgebase**

Real-time clinical reasoning support and simplified counseling instructions to empower patients and assist doctors.

## THE TECHNOLOGY

**Specialized Bengali ASR for Medical Discourse**

A tellored Automatic Speach Recognition model using speaker diarization to document native language clinical conversations.

**Azure OpenAI Processing (92% Accuracy)**

**92%**
Accuracy

High-prevision transcription utilizing GPT-so-based models to ensure clinical data integrity.

**Background Recorder.exe Utility**

A lightweight system tray utility ensures seamless audio capture during the clinical encourter.

**Parallel NER Pipeline and Redis-cached Storage**

Nine Concurrent Modules

Extracts entities from test with low latency built on FastAPI/Flask.

## THE IMPACT

**70%**
Reduction in Documentation Time

Frees up 14.8 million clinical hours annually, adding roughly 1.50 minutes of care back to every patient appointment.

**7,000–9,000 Lives Saved Annually**

By reducing prescription errors to approximately 5%, the system prevents thousands of familties caused by medication mismanagement.

**Projecto Million in Annual Economic Savings**

**Projected gainns in Annual Economic Savings**

Projected gains include $300 million in drog-related savings and $81 million in diagnostic savings through improved precision.

# PrescriptionGPT: Reimagining Our Personalized Healthcare Solutions with AI

## The Crisis: Systemic Healthcare Gaps

### The Illegibility Barrier

Poorly handwritten prescriptions mixing local and medical terms lead to frequent misinterpretation.

### High Medication Non-Adherence

Patients often self-adjust doses or stop treatment prematurely once initial symptoms improve.

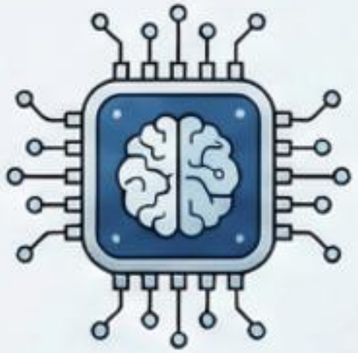
### Dangerous Drug Interactions (DDI)

A lack of point-of-care screening fails to prevent toxic combinations of drugs and food.

## THE PrescriptionGPT AI SOLUTION

### 95% OCR Accuracy Target

Advanced AI recognizes both printed and handwritton prescriptions in mixed Bengali-English formats.



| | |
|---|---|
| Image Capture to OCR: | < 2 Seconds |
| Full Prescription Analysis: | < 10 Seconds |
| Complete Report Generation: | < 15 Seconds |

**PrescriptionGPT Platform**

### Real-Time Interaction Checking

An automated engine cross-checks drugs against DGDA databases for dangerous interactions and allergies.

### 24/7 Bilingual AI Counseling

A culturally adapted conversational assistant provides dosage, diet guidance, and health literacy support.

## Projected Impacts

**40%**

**Reduction in Errors**

Target decrease in medication errors through automated safety checkers and personalized counseling.

**$500M**

**Annual Savings**

Potential household savings achieved by identifying and substituting affordable generic medications.

**1 Million**

**Target Users**

Planned user base within the first year of rollout in the Bangladesh market.

Activate Windo

# NutritionGPT: Personalized Nutrition Assistant Tailored to Culture

## Challenges

- STUNTING → Too Short for Age
- WASTING → Too thin for Age
- OBESITY → Excess Body Fat
- UNDER WEIGHT → Low Body Weight

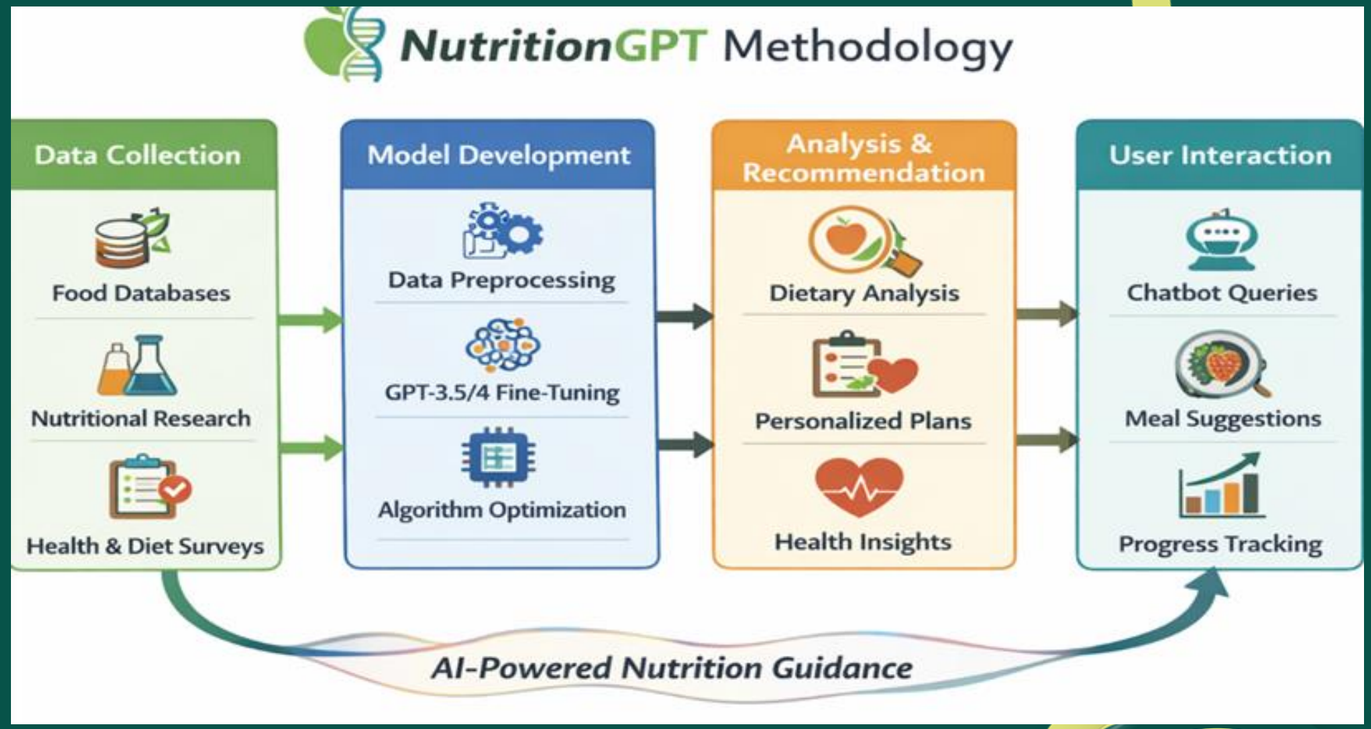
## Proposed Solution

01 Dietary Analysis

02 Meal Suggestion

03 Progress Tracking

04 Chatbot Queries




## Impact

- Accessible Nutrition Guidelines
- Healtheir Food Choice
- Prevent Diseases like Diabetes
- Increase Knowledge about Nutrition

Stay healthy
Take Healthy Food

# AI-Enabled Early Risk Prediction of Chronic Kidney Diseases

## Problem Statement

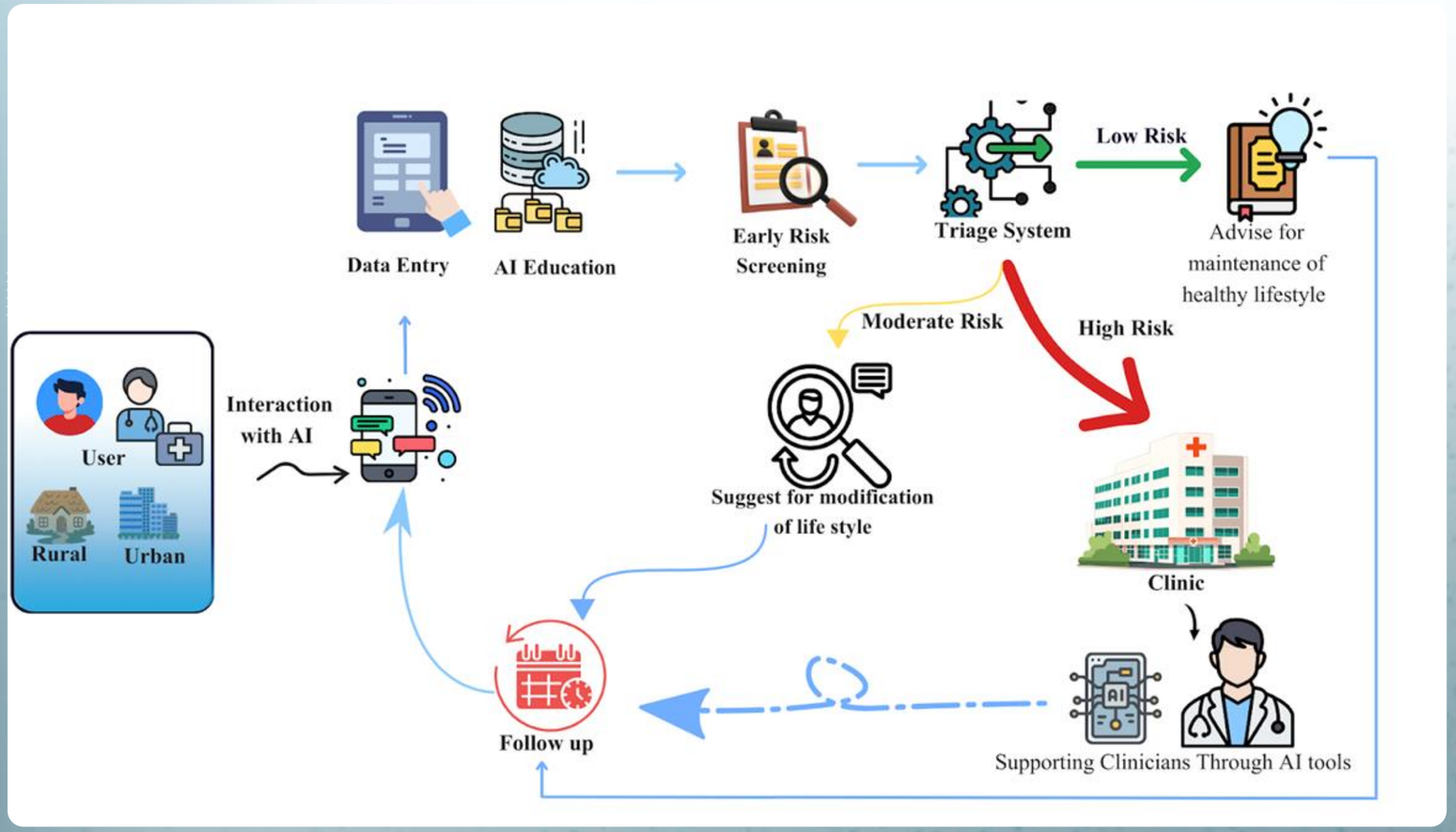
Healthcare Challenges in Bangladesh

- CKD is the 9th Leading Causes of Death in Bangladesh
- CKD diagnostics is often late in Bangladesh
- Expensive Diagnostic Barriers
- Limited Healthcare Access especially in rural area
- Lack of Continuous Monitoring ignoring lifestyle modification

## PROPOSED SOLUTION: AI-POWERED DIGITAL HEALTHCARE SYSTEM

1. AI-Powered Healthcare Platform
2. Early Risk Prediction
3. Continuous Patient Monitoring

## AI Based Early CKD Risk Prediction Echo System



# VIRTUAL HOSPITAL

A PILOT STUDY: DEVELOPMENT AND ASSESSMENT OF AN IOT-BASED INTELLIGENT HOME HEALTH MONITORING AND REFERRAL SYSTEM

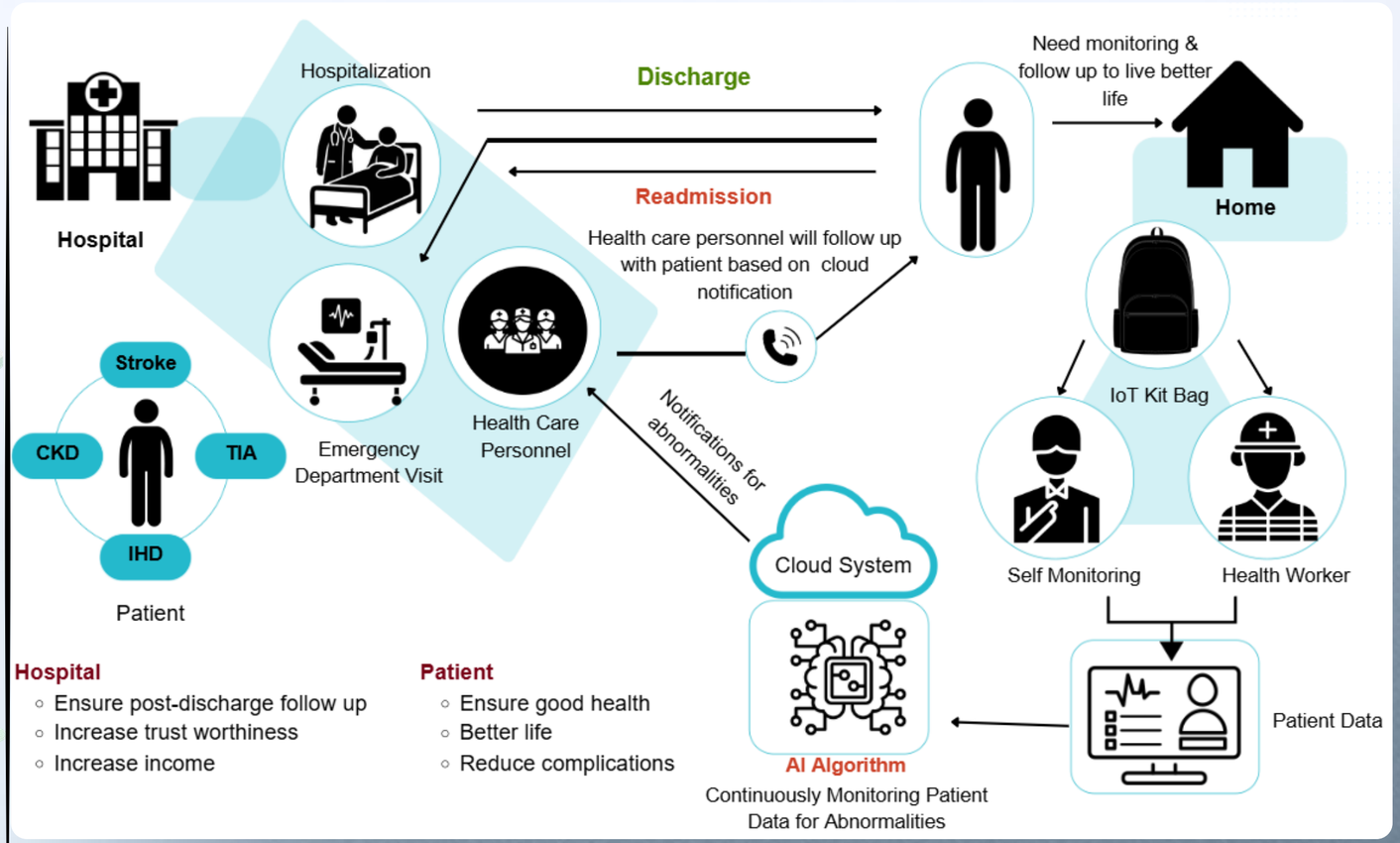
## PROBLEM STATEMENT: CHALLENGES IN POST-DISCHARGE PATIENT CARE

1. Post-discharge complications require continuous monitoring
2. Frequent hospital visits increase cost and patient burden
3. Limited hospital capacity hinders long-term follow-up
4. Rural patients lack access to specialist follow-up care
5. Current monitoring systems are fragmented and disconnected

## SOLUTION : VIRTUAL CARE PLATFORM

1. Virtual hospital platform for remote post-discharge monitoring
2. Continuous tracking of patient health data from home
3. Early detection of complications through AI-based alerts
4. Teleconsultation with doctors without hospital visits
5. Personalized care plans and follow-up recommendations

## AI and IoT Based Post Discharged Patient Care Eco System

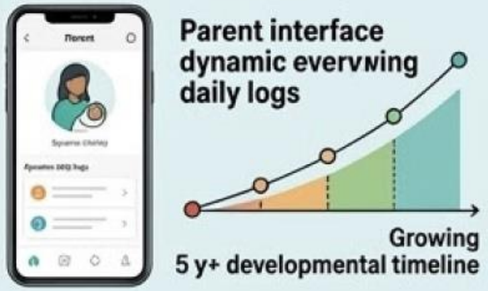


# Tiny Tracker: A Predictive Maternal and Early Childhood (0–5 Years) Neurodevelopmental Surveillance Platform Integrating Maternal Health Monitoring, Neonatal Triage, and LLM-Driven AI for Early Intervention

## The Problem
(Identifying the Care Gaps)

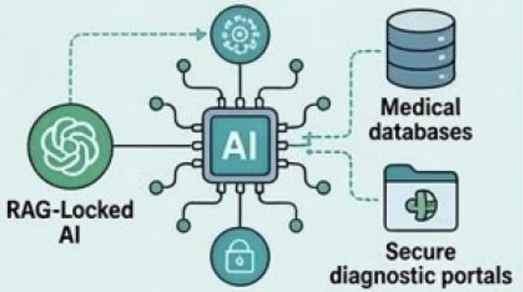
- **Early Detection Barrier:** Current tools fail to capture critical "Red Flags" during the intervals between doctor visits.
- **Maternal & Neonatal Risk:** Critical windows for Preeclampsia, GDM, and Newborn Sepsis go unmonitored at home.
- **The "Wait and See" Failure:** Delayed neurodevelopmental (NDD) referrals cause missed windows for critical early intervention.
- **Geographic Barrier:** Access to pediatric specialists is concentrated in urban hubs, leaving rural families isolated.

## The Solution
(How Tiny Tracker Works)




- **Maternal/Neonatal Logic Gate (The Sentry):** Coded clinical standards (BP, Glucose, Temp) to instantly detect Acute Risks.
- **0–5 Dual-Track surveillance:** Checks 300+ milestones via the "75% Clinical Floor Rule" (Track A) while simultaneously observing behavioral clusters (Track B).
- **The "Wait and See" Killer:** Multimodal data capture, including biometric sensors and Video Verification of high-stakes milestones, provides objective evidence.
- **The Adaptive Care Path:** Platform recalibrates post-diagnosis, switching to Personal Potential Curves to track therapy efficacy.

## The Technology
(The Predictive Infrastructure)




- **RAG-Locked AI Assistant LLM:** AI provides advice by querying verified medical databases, ensuring 0% hallucination and total reliability.
- **Computer Vision (Video Analysis):** Uses side-by-side comparison logic to help parents distinguish typical from delayed motor behaviors.
- **OCR Clinical Intake:** Optical Character Recognition automatically extracts diagnosis codes and recommendations from uploaded specialist reports.
- **Advanced Pattern Engines:** Proprietary algorithms that sync joint attention logs and regulatory clusters to detect ASD/ADHD patterns.
- **HL7/FHIR Interoperability:** Standardized medical data exchange for secure, one-click sharing with hospital EMR systems.
- **Trend Analysis Engine:** Uses Z-score variations to visualize growth in non-linear pathways for post-diagnosis progress.

## The Impact
(Transforming Care Outcomes)

- **Predictive Early Intervention:** Enables objective referrals early on, bypassing the "Wait and See" delay.
- **Specialist Efficiency:** "Doctor-Ready" PDF summaries reduce history-taking time, enabling specialists to make faster, data-driven decisions.
- **Evidence-Based NDD Research:** Longitudinal JSON data bundles contribute to global research on neurodevelopmental pattern detection.
- **Scalable Care Model:** Sustainable SaaS platform designed to deliver clinical-grade surveillance to urban and underserved rural regions.

# Bolte Chai+: A Tech-Driven Communication Platform for Non-Verbal Children Integrating Cognitive Research, Artificial Intelligence (AI), Augmented Reality (AR), and Virtual Reality (VR) to Enhance Speech Therapy Accessibility in Bangladesh

## The Problem

**700,000+**

children face challenges in expressing themselves verbally

**The "One-Size-Fits-All" failure**

Curent tools are English-centric, fail to adapt to unique needs

vs

**Geographic Barrier**

Therapy concentrated in cities, leaving 70% in rural areas without support

## The Solution

**How Bolte Chai+ works**

**Multimodal Engagement**

Personalized activities with interactive cues and immersive simulations

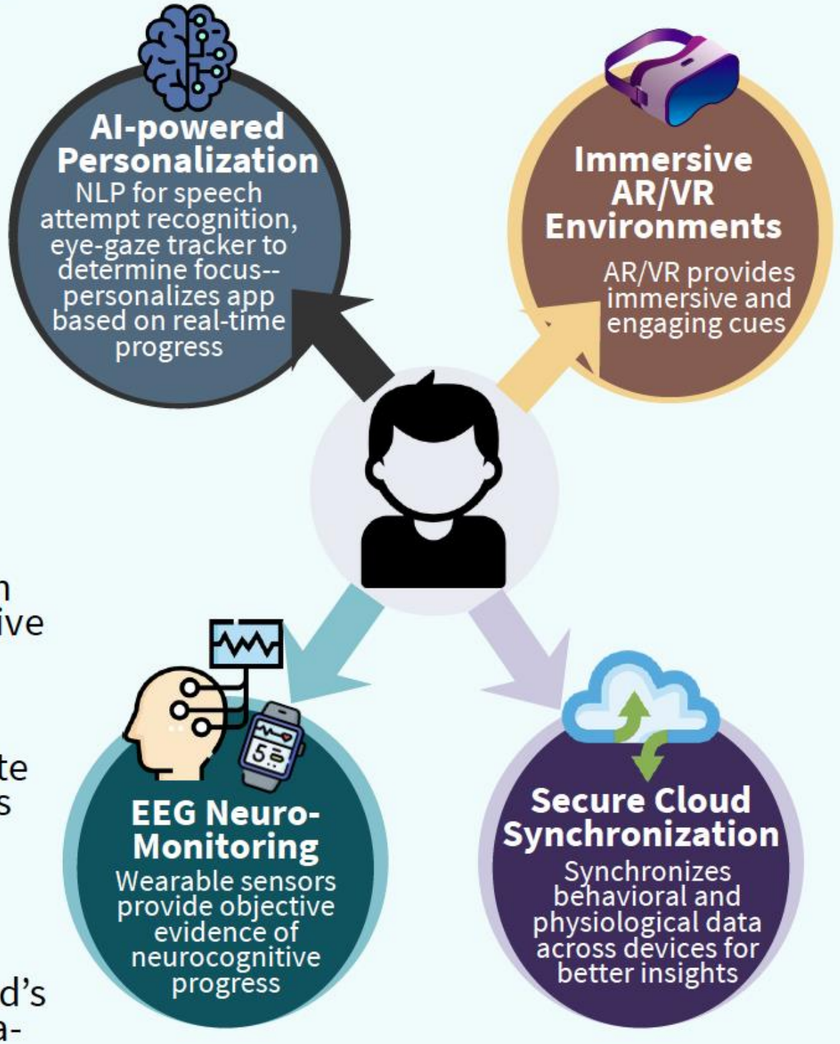
**Progress Reports**

Intelligent analytics translate behavioral data for parents and therapists

**Closed-Loop Personalization**

The app adapts to each child's cognitive needs using data-driven insights

## The Technology




## The Impact

**Improving Quality of Life**

Enhances independence and social inclusion; reduces caregiver burden

**Advancing Global Research**

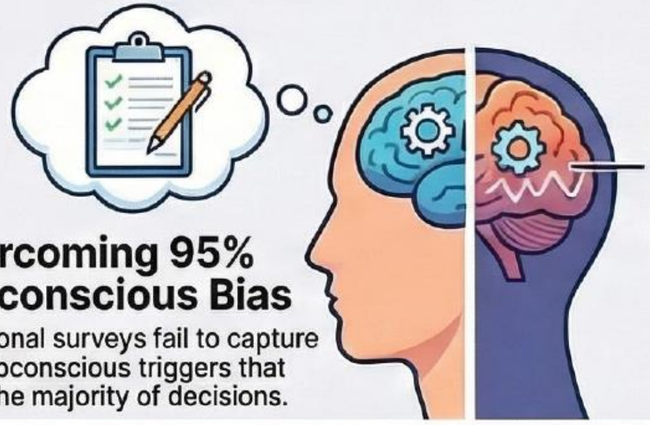
Contributing to global autism and neuro-engineering knowledge

**Scalable and Sustainable**

Cost-effective SaaS model ensures long-term sustainability across urban and rural areas

# **Transformative Consumer Neuroscience:** The Neuromarketing **BCI Platform**

## PROBLEM: THE 'SAY-DO' GAP

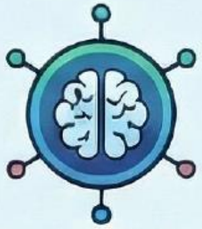


Creates a "say-do" gap in consumer behavior analysis.

### Overcoming 95% Subconscious Bias

Traditional surveys fail to capture the subconscious triggers that drive the majority of decisions.

## SOLUTION: INTEGRATED & INDIGENOUS PLATFORM



Neuromarketing BCI Platform

### Multimodal Data Integration

Synchronizes BCI brain activity with eye-tracking to eliminate fragmented, siloed research data.

### Indigenous Research Solution

Replaces expensive foreign tech with algorithms specifically tailored to the Bangladeshi cultural landscape.

## TECHNOLOGY: CORE & INSIGHTS

### Immersive AR/VR Simulations

Recreates realistic 3D retail environments to measure real-world shopper reactions to packaging.

MINDWAVE ANALYTICS
Marketing In The Brain

### Data Table

| BCI (EEG) | Eye-tracking | Implicit Association (IAT) |
|---|---|---|
| Frontal Alpha Asymmetry | Visual Fixation/ROI | Reaction Time Latency |
| Subconscious motivation and pricing comfort | Real-time attention and advertisement clarity | Hidden brand blases and attitudes |

## IMPACT: NATIONAL & COMMERCIAL

**10+ local companies already engaged**

Industry leaders are already participating in demos, validating the platform's commercial viability.

## Building a 4IR-Ready Economy

**Knowledge Hub** (Scopus-indexed publications)

**Patents**

**High-Value Neuro-analytics Jobs**

# NEUROPILOT

## An Intelligent EdTech Framework for Cognitive Engagement & Educational Equity

## Current Problems

**Digital learning suffers from rapidly declining attention spans (often <10 mins).**

Urban-Rural Divide

Difference In Teaching Expertise

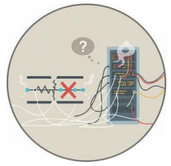
Expensive Private Tutors For Monitoring

External Distraction And Content-Induced Boredom

### Gaps in Research

Complex EEG Headbands

High Cost



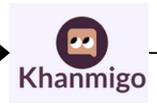
No Feasibility in Real-time

### Gaps in Industry

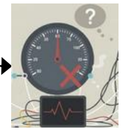
Competitors

Static Content

Learning Issues

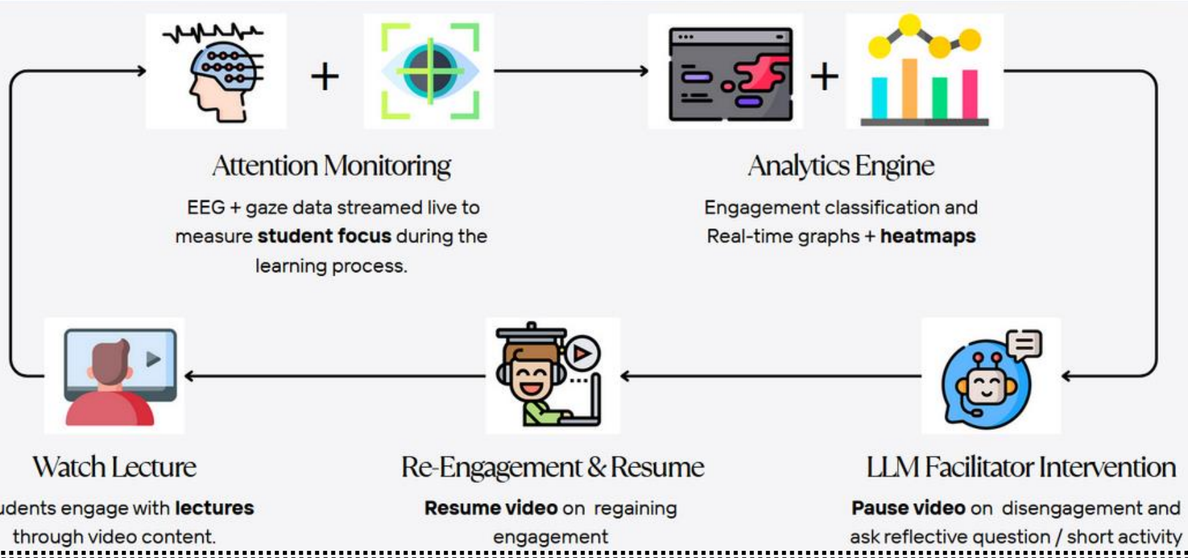



State-of-the-art



Lack of Monitoring

## Solution

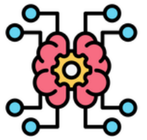



## Technologies

Brain-Computer Interface

Agentic AI Based Facilitator

Eye-Gaze Tracking



Artificial Intelligence

| Feature | Traditional EdTech | The NeuroPilot Shift |
|---|---|---|
| Measurement | Performance proxies (clicks, quiz, scores) | Neurological ground truth (EEG & Gaze) |
| Intervention | Reactive (waits for failure) | Proactive (intervenes mid-distraction) |
| AI Role | Open-loop (waits to be asked) | Closed-Loop (Autonomous adaptiation) |

## Outcomes

Multimodal data from 200+ participants

Validated SOTA Engagement Classification Model

Real-time Prototype With Engagement Monitoring and Facilitation

## Impact

Equal Talent, Equal Hardwork, Equal Achievement

Better Learning Environment (SDG 4)

Building Capable Gen-Z Workforce Contributing to Economic Growth

# Multimodal Identification of Educational Content Attractiveness

Moving beyond self-reports to capture objective cognitive and emotional engagement.

## THE PROBLEM

**Social Desirability Bias:** Compromised by societal pressure & retrospective flaws.

**The Attention Fallacy of LMS Logs:** Tracks physical activity, fails to verify cognitive presence.

**The Critical Gap:** Studies measure Student Engagement (the result) rather than Content Attractiveness (the cause).

## THE SOLUTION

**Content Attractiveness**

=

**Cognitive Resonance** (EEG)

+

**Perceptual Fluency** (Eye Tracker)

+

**Affective Connection** (Facial Expression & PPG)

## THE TECHNOLOGY

**Multi-Sensor Array (Scannable List)**

- EEG Sensor
- Eye Tracker
- Webcam for Facial Expression
- PPG Sensor

## IMPACT

**Pedagogical Optimization:** Real-time, data-driven curricula adjustments based on objective student response.

**Enhanced Learning Outcomes:** Reducing cognitive overload and "boredom-induced dropout".

**Standardized Benchmarking:** A universal, empirical framework for evaluating global materials.

# DEEP-DEPRESSION: An AI-Powered EEG-Guided Brain–Computer Interface (BCI) Enabled Solution for Objective Depression Diagnosis to Strengthen Mental Health Care in Low Resource Settings

Bridging the mental health diagnostic gap in low-resource settings through objective AI-powered brain signal analysis.

## THE MENTAL HEALTH CRISIS IN BANGLADESH

**18.7%** Adults &
**12.6%** Children
Suffer from Depression

**91%**
**RECEIVE NO TREATMENT**
Massive Diagnostic Gap

During the pandemic, Depression rates in Bangladesh rose nearly **6 to 7 times**

**75%** of individuals injured during July protests suffer from depression

**Subjective Diagnosis Creates a**
**3.2-YEAR DELAY:**
Prone to recall arror, observer bias, social stigma.

## THE PROPOSED SOLUTION: OBJECTIVE NEUROPHYSIOLOGICAL INSIGHT

**OBJECTIVE EVIDENCE VIA BCI, EEG, & AI:**
Brain Signals Cannot Lie

BRAIN SIGNALS REVEAL HOW DEPRESSION ALTERS NEURAL CONNECTIVITY AND ELECTRICAL ACTIVITY

EEG PROVIDES A SIMPLE, LOW-COST, AND PATIENT-FRIENDLY METHOD TO RECORD BRAIN ACTIVITY LINKED TO DEPRESSION

EEG-BASED BCI SYSTEM CAN TRANSLATE BRAIN ACTIVITY INTO MEASURABLE PATTERNS, PROVIDING OBJECTIVE EVIDENCE.

## TECHNOLOGY PIPELINE: FROM RAW SIGNAL TO DIAGNOSIS

**STEP 1: Raw EEG Signal Acquisition**
Records resting states & Emotional Stroop tasks.

**STEP 2: Signal Preprocessing**
Filtering, Artifact Removal, Common Average Referencing.

**STEP 3: Feature Extraction and Selection**
Time/Frequency domains, Recursive Feature Elimination.

**STEP 4: Pattern Classification**
Support Vector Machine (SVM) Model.

**STEP 5: Expert Review & Diagnostic Report**
Ai generates indicators for Psychiatrist's Clinical Decision.

## WHY THIS PROJECT MATTERS

**90.5%**
**90.5% ACCURACY IN DETECTING DEPRESSION**

**ENABLES OBJECTIVE, CLINICIAN-SUPERVISED MENTAL HEALTH ASSESSMENT IN RESOURCE-LIMITED SYSTEMS.**

# ADVANCING FORENSIC INTEGRITY IN BANGLADESH

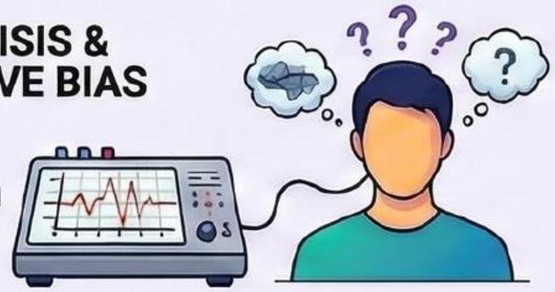
## A Multimodal Deception Detection Framework

## CURRENT PROBLEMS

**200,000+**
**CYBERCRIME COMPLAINTS**

### TRUST CRISIS & SUBJECTIVE BIAS

Traditional polygraph polygraph tests are compromised compromised methods.



### OPERATIONAL BOTTLENECKS

Manual verification, unable to keep pace with a explosion of modern digital fraud and interconnected fraud networks.

## THE SOLUTION

**AI-Driven Automation**

Transitioning from human-based, subjective interrogation to an objective, AI-driven framework.

**Biological Truth**

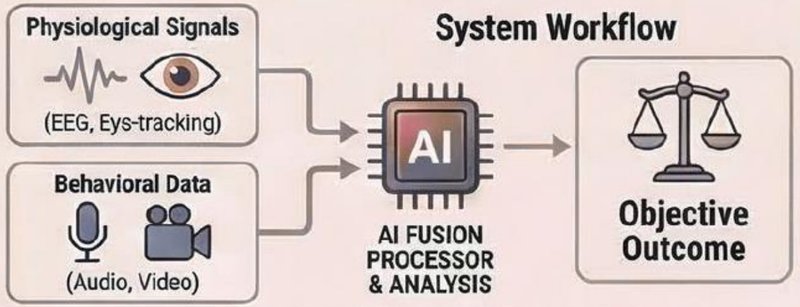
Monitors objective involuntary brain signals & cognitive load

**Culturally Valid Framework**

Tailored to unique linguistic, behavioral, and cultural nuances of Bangladeshi population

## TECHNOLOGIES

**System Workflow**



### SCALABLE FORENSIC HARDWARE

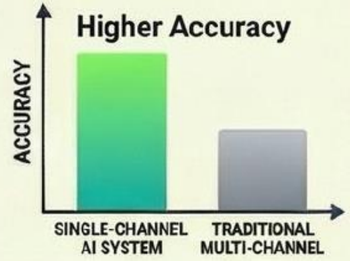
Uses affordable, single-channel EEG technology designed for local law enforcement deployment.

## OUTCOMES

### FIRST BANGLADESHI DECEPTION DATASET

Successfuly creation of a specialized demographic milestone for accurate local forensic modeling.

### SUPERIOR HIGH-FIDELITY CLASSIFICATION




## IMPACT

### HUMAN RIGHTS & ETHICS

Reduced reliance on physical coercion and upholding constitutional rights.

### POLICY REFORM & ADMISSIBILITY

Establishing a framework for neuro-evidence as admissible proof in Bangladeshi courts.

NotebookLM

# Smart Governance for All: The AI-Driven Voice Kiosk

## THE ACCESSIBILITY GAP

**The Literacy & Language Barrier**

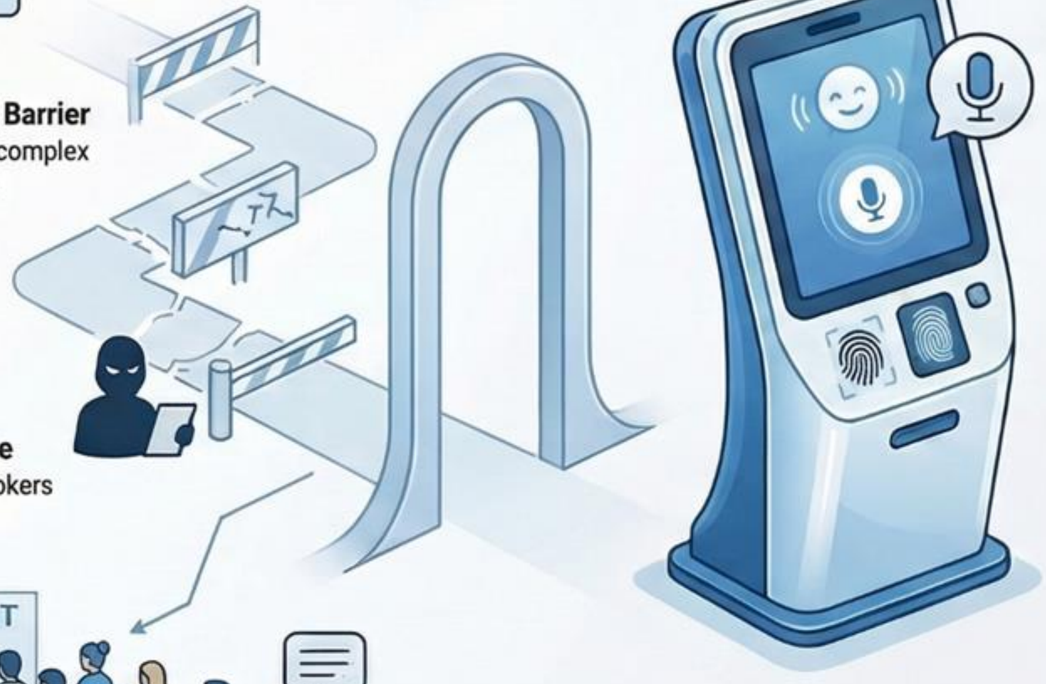
Citizens often struggle with complex digital interfaces and formal bureaucratic language.

**Intermediary Dependence**

High rellance on informal brokers increases service costs and reduces transparency.

**Operational Overload**

Public institutions face overcrowded counters and repetitive manual inquiries.



## THE AI-POWERED KIOSK

**Multilingual Voice Interaction**

Supports Bangla, English, and regional dialects (like Sylheti) for low-literacy users.

**Secure Biometric Authentication**

Integrates National ID (NID) and fingerprint login for personalized, secure access.

**Generative AI Automation**

Converts speech to forms, simplifies circulars, and drafts formal complaints.

## MEASURED PUBLIC IMPACT

**Reduced Processing Time**

AI automation decreases manual workload and average inquiry handling time.

**Direct Financial Savings**

Transparency tools and direct access reduce travel costs and intermediary payments.

**Inclusive Digital Inclusion**

Specifically designed with elderly-friendly speech modes and disaster guidance support.

# SmartReceptionist: Redefining Front-Desk Intelligence for Bangladesh 2.0

## THE CHALLENGE:
LIMITATIONS OF MANUAL FRONT DESKS

### Antiquated Manual Processes
Traditional desks rely on human-only staffing, leading to high manpower costs, limited service hours, and operational bottlenecks.

### The Linguistic & Digital Divide
Current manual systems struggle to support the complex flus of Bangla, English, and 'Banglish' code-switching used by the modern Gen Z population.

### Security & Access Friction
Lack of secure, contactless authentication in high-traffic environments like hospitals and banks leads to slower check-ins and safety conneerns.

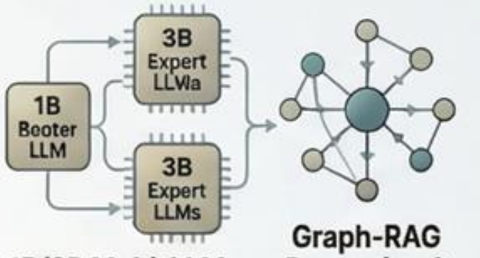
## THE SOLUTION:
AN OMNICHANNEL ECOSYSTEM

### Bangladesh's First Autonomous Robotic Receptionist
A pioneering robotics system designed for 24/7 service, moving beyond passive kiosks to active, human-like interaction.

### Omnichannel Service Delivery
Operates as a unified platform across physical robotic kiosks, web-based administrative dashboards, and mobile integration.

### Bangladesh 2.0 Vision
A commercially viable blueprint for service automation that fosters local high-tech startups and national technological sovereignty.

## THE TECHNOLOGY:
ADVANCED MULTI-TIER AI STACK




### 1B/3B Multi-LLM Cognition Engine
A lightweight 18 "romet" LLM classifies intent and directs quaries to 30 "espert" LLMs for high-prevision, contextual response generation.



### Graph-RAG Reasoning & Knowledge
Employs a Graph-based Retnosal-Augmented Generation engine for multi-hop logical traversal and fact-grounded responses.

### Multimodal Biometric Layer
Utilizes ArsFace facial recognition and ECAPA-TDNN speaker recognition for secure, touchless identity verification.

### Fine-Tuned Perception & Output
Features a Whisper-+3-turbe model optimized for regional dialeets and a Sesame-based TTS for expressive, natural valve synthesis.

## STRATEGIC IMPACT

### 24/7 Operational Efficiency
Eliminates downtime by providing round-the-clock service without the need for additional front-desk manpower.

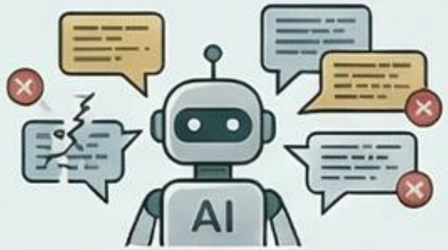
### Contactless & Secure Interaction
Enhances institutional security through advanced face and voice authentication, ideal for modern, high-security service environments.

### Scalable Automation Infrastructure
Provides a modular B2B/B2G tramework that can be easily deployed across healthcare, banking, and government sectors.

# IslamicGPT: Bridging the Digital Faith Gap with Verifiable Intelligence

## THE DIGITAL FAITH GAP

### The Hallucination Crisis

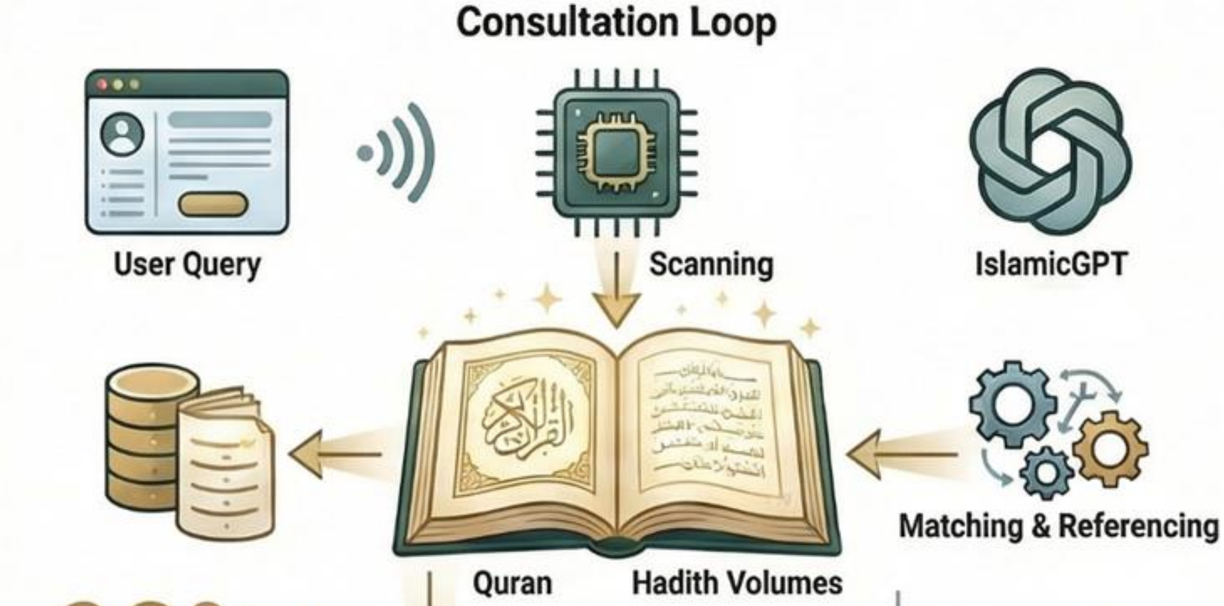



Standard AI often generates unverified content, fake fatwas, and inaccurate religious interpretations.

### Fragmented Discourse

Gen-Z Muslims encounter extremist narratives and digital noise in unmoderated online spaces.

## THE 'DIGITAL SCHOLAR' APPROACH




**90%+**

**Accuracy Rate**

Pilot tests show significant performance superiority over standard Large Language Models

**Digital Seal of Authenticity**

Every response includes verifiable citations and a seal from authorized religious bodies

## 2050 VISION & IMPACT

### Promoting Social Cohesion

Acts as a shield against religious misinformation to prevent digital polarization.

### Universal Digital Access

Provides scholarly expertise to all citizens via mobile app and website portals.

# LegalGPT: Bridging the Justice Gap in Bangladesh

## THE PROBLEM: AN OVERWHELMED SYSTEM & UNSAFE AI

**4.65 MILLION**
**Pending Cases**

Creates a massive "First Mile Bottleneck," with ~4 million cases stuck in lower courts

### Justice is Inaccessible

A high logal literacy gap and poverty prevent citizens from navigating the system.

### The Core Challenge: Legal Hallucinations

General AI models are unsafe, frequently inventing false citations and providing dangerous answers.

## THE SOLUTION: A GROUNDED "FIRST-MILE" LEGAL ASSISTANT

Safety-by-Design

### Evidence-Based & Safe Architecture

RAG (Retrieval-Augmented Generation)

### Grounded in Authoritative Legal Texts

### Citizen Service Flow

**1. EXPLAIN**
Provides information in plain language.

**2. TRIAGE**
Escalates users to human support.

**3. REFER**
Connects to NLASO for human support.

### Accessibility for the Grassroots

Voice-First interface

**Plain Language Design**
Serves lower-literacy populations.

# AI-Assisted Judicial Decision Support System

## Tackling Case Backlog in Bangladesh

### Problem

- **122,422** Drug Cases Filed (2020–2025)
- Only **13,376** Cases Resolved
- Manual Processing of Case Documents
- Dependence on Physical Files
- Slow Judgment Drafting Process

Result: *Severe judicial backlog and debyed justice*

### AI-Based Solution

**AI-assisted system** for Judges

- Automatic summarization of evidence
- Identification of relevant laws & precedents
- Generation of structured draft judgments
- Digital case document management

*Judge makes the final decision*

### Technology

- **Large** Language Models (LLMs)
- **Retrieval-Augmented** *Generation* (RAG)
- Natural Language Processing (NLP)
- **Secure** cloud-based *legal platform*

### Impact

- **Faster** Judgment Drafting
- **Reduced** Case Backlog
- Greater Transparency & Accountability
- **Improved** Access to Justice

*Supports **Smart Bangladesh** digital judiciary vision.*

# Application of Artificial Intelligence in Writing Judicial Pronouncement to Tackle the Backlog of Cases.

## Problem

1. 122,422 Drug Cases Filed (2020–2025)
2. Only 13,376 cases resolved
3. Manual processing of case documents
4. Dependence on physical files
5. Slow judgment drafting process

**Result: Severe judicial backlog and delayed justice**

## AI-Based Solution

1. AI-assisted decision support system for judges
2. Automated evidence summarization
3. Legal precedent & law identification
4. Structured draft judgment generation
5. Digital case management platform

**Human judge makes the final decision**

## Technology

1. Large Language Models (LLMs)
2. Retrieval-Augmented Generation (RAG)
3. Natural Language Processing (NLP)
4. Secure cloud-based legal infrastructure

## Impact

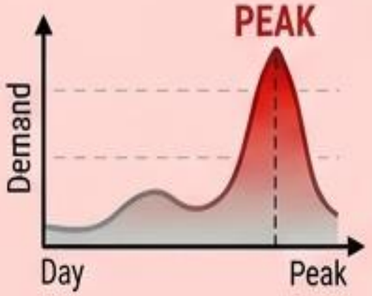
1. Faster judgment drafting
2. Reduced case backlog
3. Greater transparency & accountability
4. Improved access to justice

**Supports Smart Bangladesh Digital Judiciary**

# iPower

Intelligent Electricity Demand Forecasting & Optimization Platform for Bangladesh

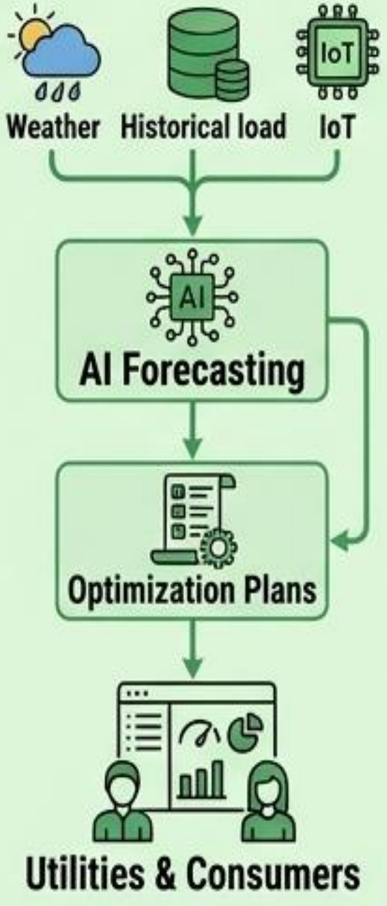
## The Problem




- **Population: 175M+**
- Rapid economic growth straining resources
- **Chronic Supply-Demand Mismatch**

- **High Cost of Fossil Fuel Imports**
- **High Operational Costs**

## Proposed Solution




- Forecast Automation, Cloud
- Real-time Forecasts
- Forecast (Consumer to
- National Level
- Cost Savings **~200 crore BDT**
- Reduced Blackouts

## Proposed Technology

**AI/ML**
- LSTM
- Transformers
- Ensembles
- Concept Drift Detection

**Data**
- Pandas
- Scikit-learn
- TensorFlow

**Big Data**
- Spark
- MongoDB
- PostgreSQL

**Big Data**
- Hadoop
- Spark

**Databases**
- MongoDB

**Cloud**
- AWS/Azure

**Other**
- XAI (SHAP/LIME)
- Federated Learning
- IoT

# What's Next | Industry Ready

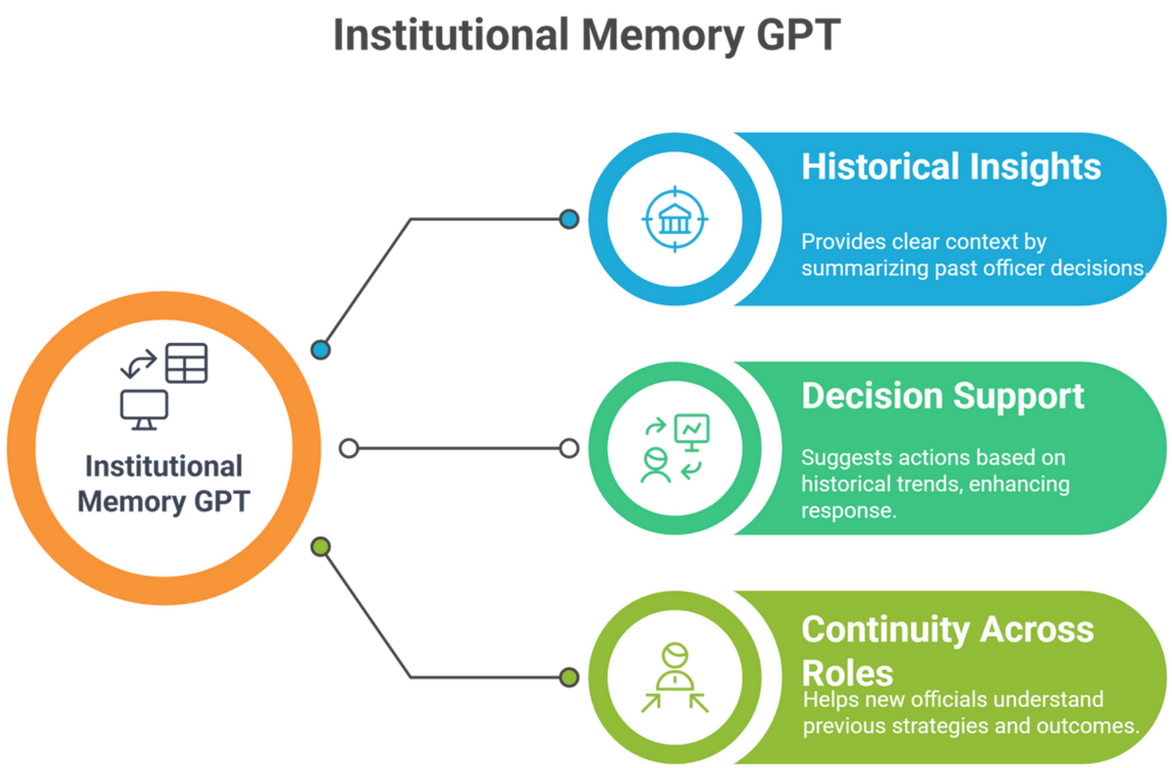
## An AI-powered personalized career pathway recommender based on cognitive, personality and psychometric assessments

Cognitive Ability

Personality Traits

Psychrometric Profiles

# AI for Governance
# Institutional Memory GPT




## Benefits:

Reduces delays caused by knowledge gaps during transitions of people.

Promotes consistency in governance by preserving institutional memory.

## Real-Life Example:

An officer handling disaster relief can instantly access prior resource allocation decisions and outcomes, ensuring consistency and efficiency.

# Leveraging AI in Government and Private Sectors

**Objective**

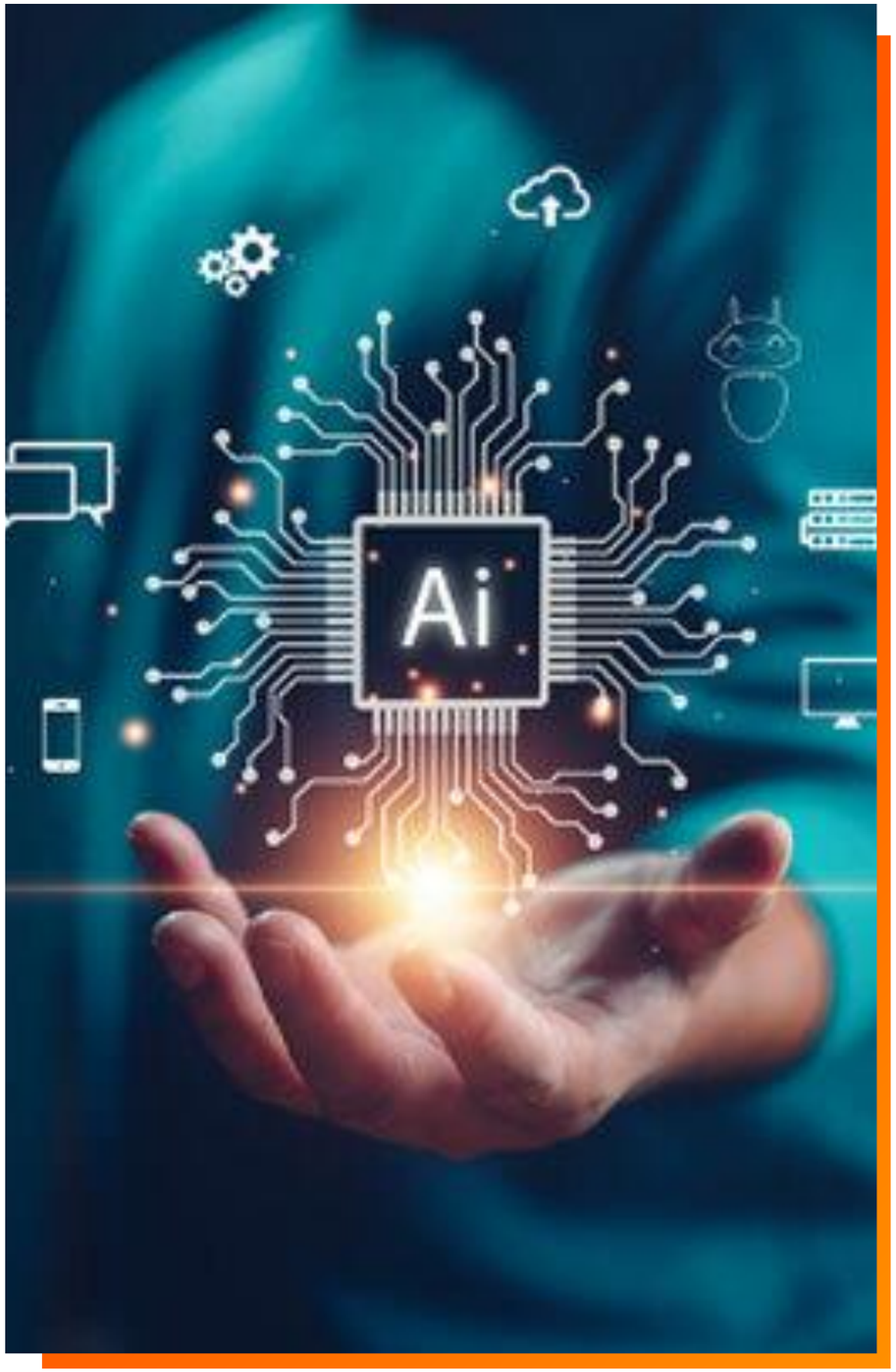
Enhance efficiency, decision-making, and institutional memory.

**Functions:**

- Data-Driven Decision-Making
- Regulation Formulation
- Addressing Emerging Challenges
- Crisis Management

**Implementation Steps:**

- Analyze current processes to identify AI opportunities.
- Build infrastructure and train staff.
- Pilot, refine, and scale AI solutions



# Innovation Hub in Govt. Offices and Large Corporates

**Objective**

Innovation hub in government offices can facilitate adaptation to new technologies, foster innovation to create effective regulations, and support the implementation and utilization of emerging ideas and innovations.

**Functions:**

- Partnerships with universities and research institutions.
- Pilot emerging technologies in real-world scenarios.
- Develop services tailored to public needs.

**Implementation Steps:**

- Form cross-functional teams with experts.
- Set up a framework for testing and scaling ideas.
- Collaborate with stakeholders to implement innovations.

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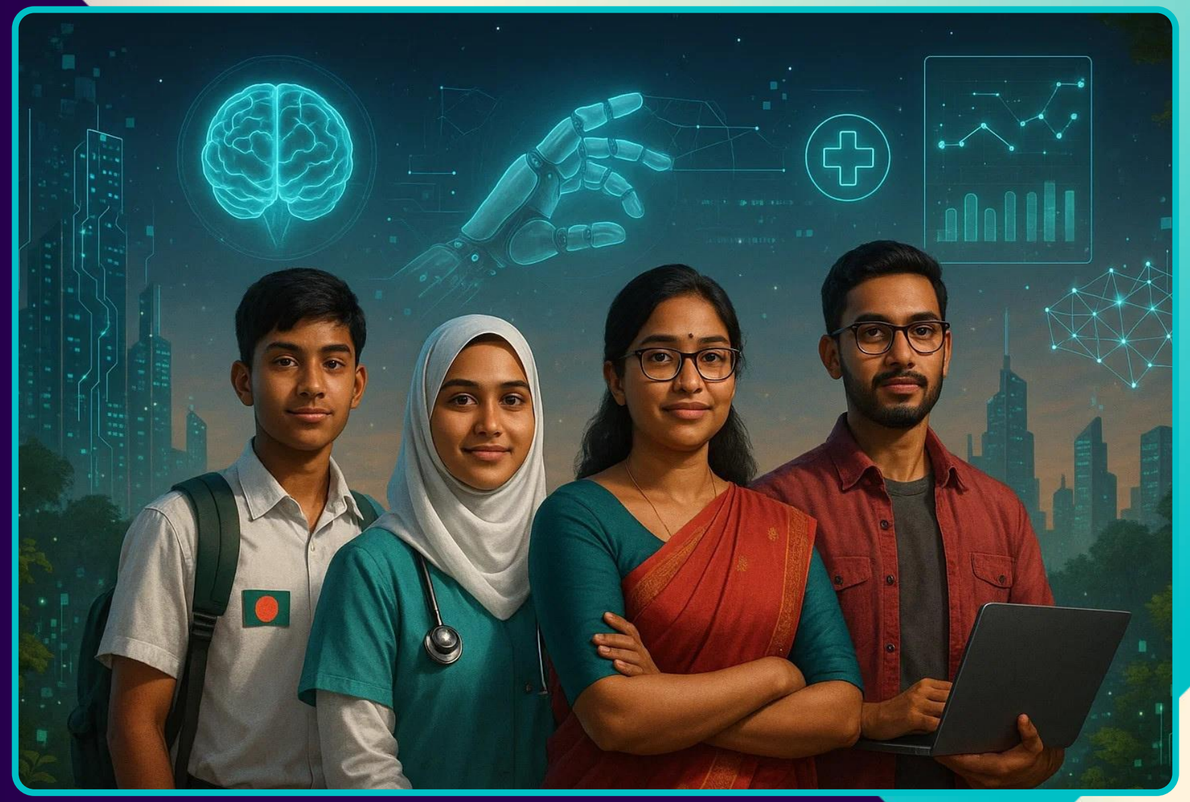
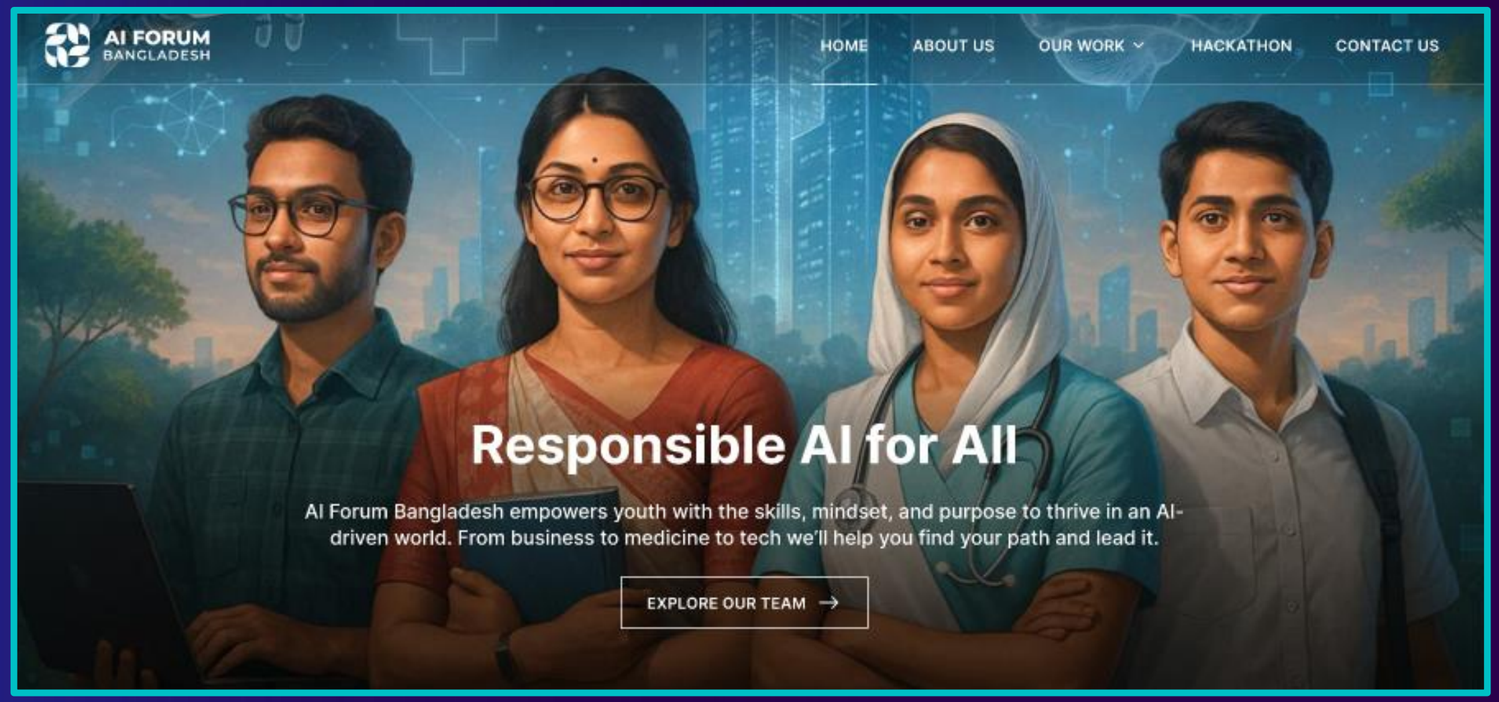
UNITED INTERNATIONAL UNIVERSITY



Responsible AI for All

# AI FORUM BANGLADESH

# AI FORUM BANGLADESH




**Acts as a national platform to drive AI adoption, innovation, and responsible practices**

**Engages students, professionals, academia, industry, and policymakers**

**Provides a bridge between learning, research, and career growth.**

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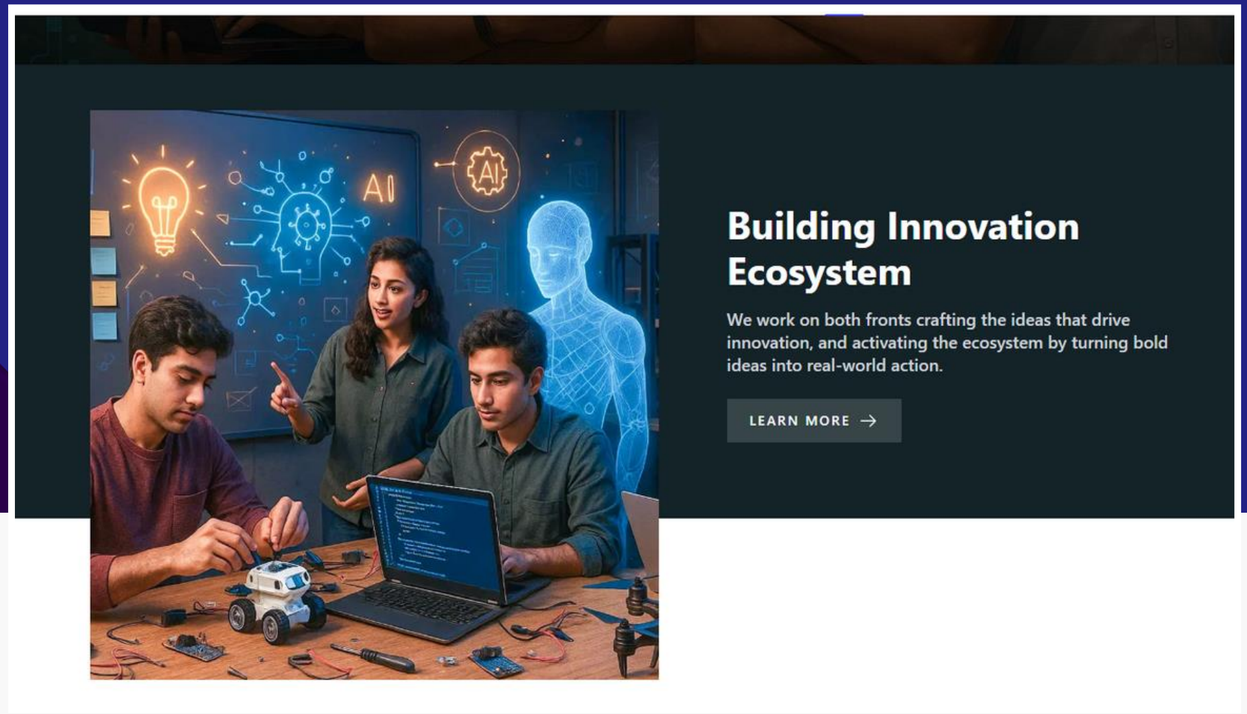
BUILDING INNOVATION ECOSYSTEM




# Building Innovation Ecosystem

A national platform for fostering culture of innovation and ecosystem growth through AI-driven initiatives and collaborations

AI ACADEMY

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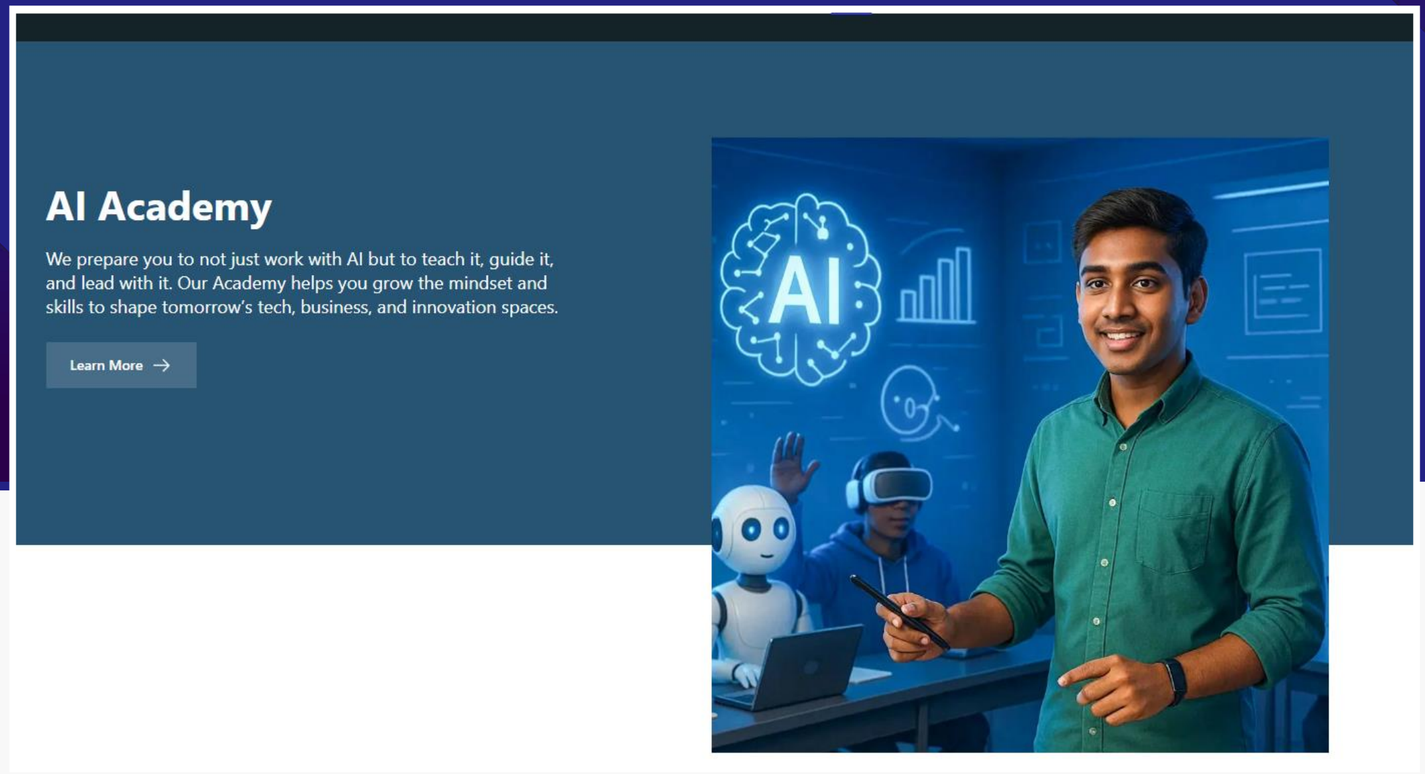


# AI Academy

A knowledge center for AI training and upskilling

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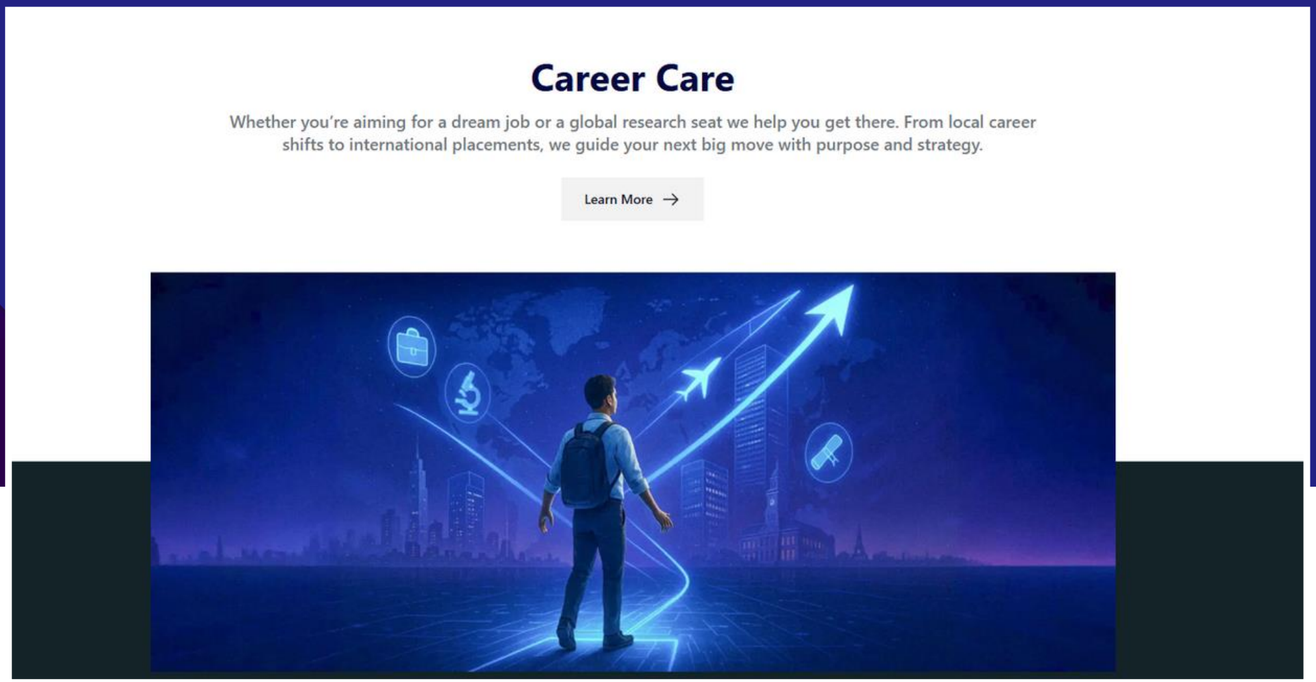
Career Care




# Career Care

A personalized career support center offering mentorship, counselling, placement, and professional growth opportunities for local and global markets

# Opportunities : What We Can Achieve

**150** Universities
**150** Innovation Hubs
**150** Innovation Leaders

**1** Innovation Hub Solves
**10** Societal and Industrial Problems

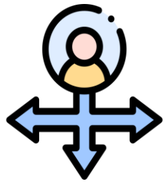
**150** Innovation Hubs will solve
**1500** Problems

# Call to Action

Collaboration

Partnership

Support



Strategic Action for Co-creation & Scale-Up

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# Thank You!